ELE-plan for i dag

Lidt repetition ?



DC Bias

Offsetspændingen V_{off}

- *Årsag*
- *Placering i ækvivalentdiagram*
- *Datablad for µA741*

Små opgaver

Biasstrøm I_B og Offsetstrøm I_{off}

- *Årsag*
- *Placering i ækvivalentdiagram*
- *Hvordan opgives disse i datablade?*

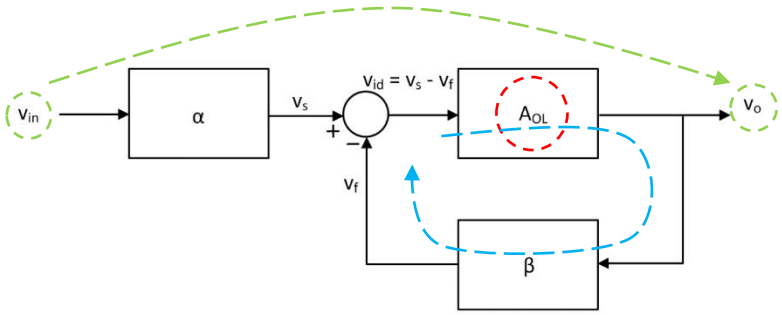
Midtvejsdiskussion

D5

Offsetjustering

- *Intern/ekstern justering*

Potentiometre og Fysisk placering af AD590

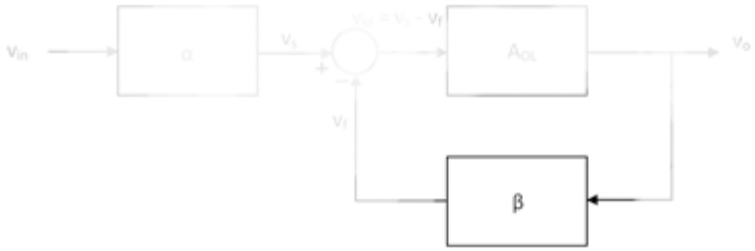


A_{OL}: ***Åben**sløjfeforstærkningen (A)* → $A_{OL} = \frac{v_o}{v_{id}}$

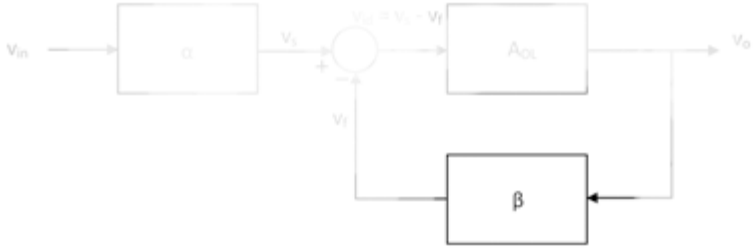
A_{CL}: ***Lukket**sløjfeforstærkningen (A_f)* → $A_{CL} = \frac{v_o}{v_{in}}$

βA_{OL}: ***Sløjfe**forstærkningen*

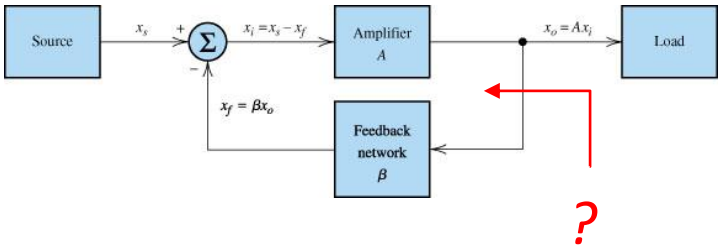
Tilbagekoblingen β



Tilbagekoblingen β



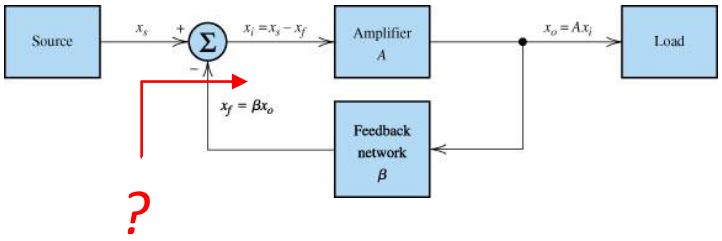
*Hvordan skal kredsløbet se ud som **kilde**?*



Set fra R_L

Strøm
Spænding

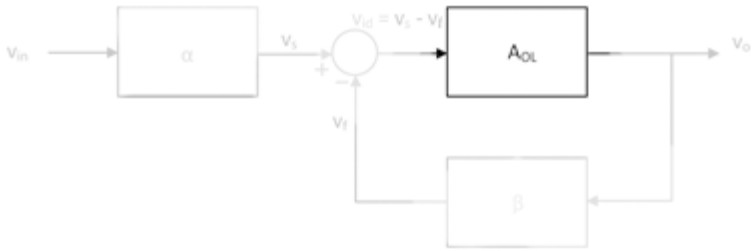
*Hvordan skal kredsløbet se ud som **belastning**?*



Set fra v_s

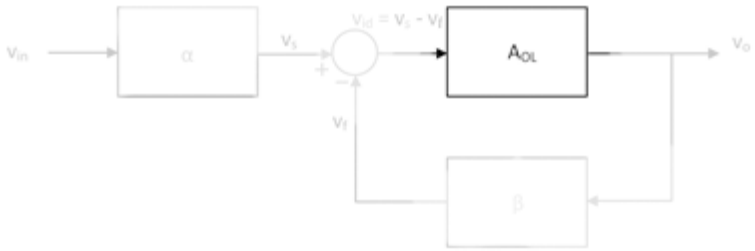
Serie-
Parallelt-

Åbensløjfeforstærkningen A_{OL}



$$A_{OL}(f) = \frac{A_{0OL}}{1+j\frac{f}{f_{BOL}}}$$

Åbensløjfeforstærkningen A_{OL}



$$A_{OL}(f) = \frac{A_{0OL}}{1 + j\frac{f}{f_{BOL}}}$$

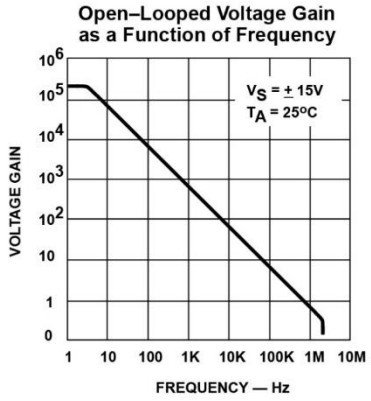
A_{OL} frekvensafhængig

↓

K_f frekvensafhængig: $$K_f = \frac{1}{1 + \frac{1}{\beta \cdot A_{OL}}}$$

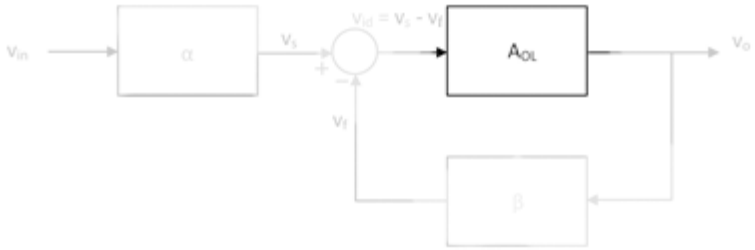
↓

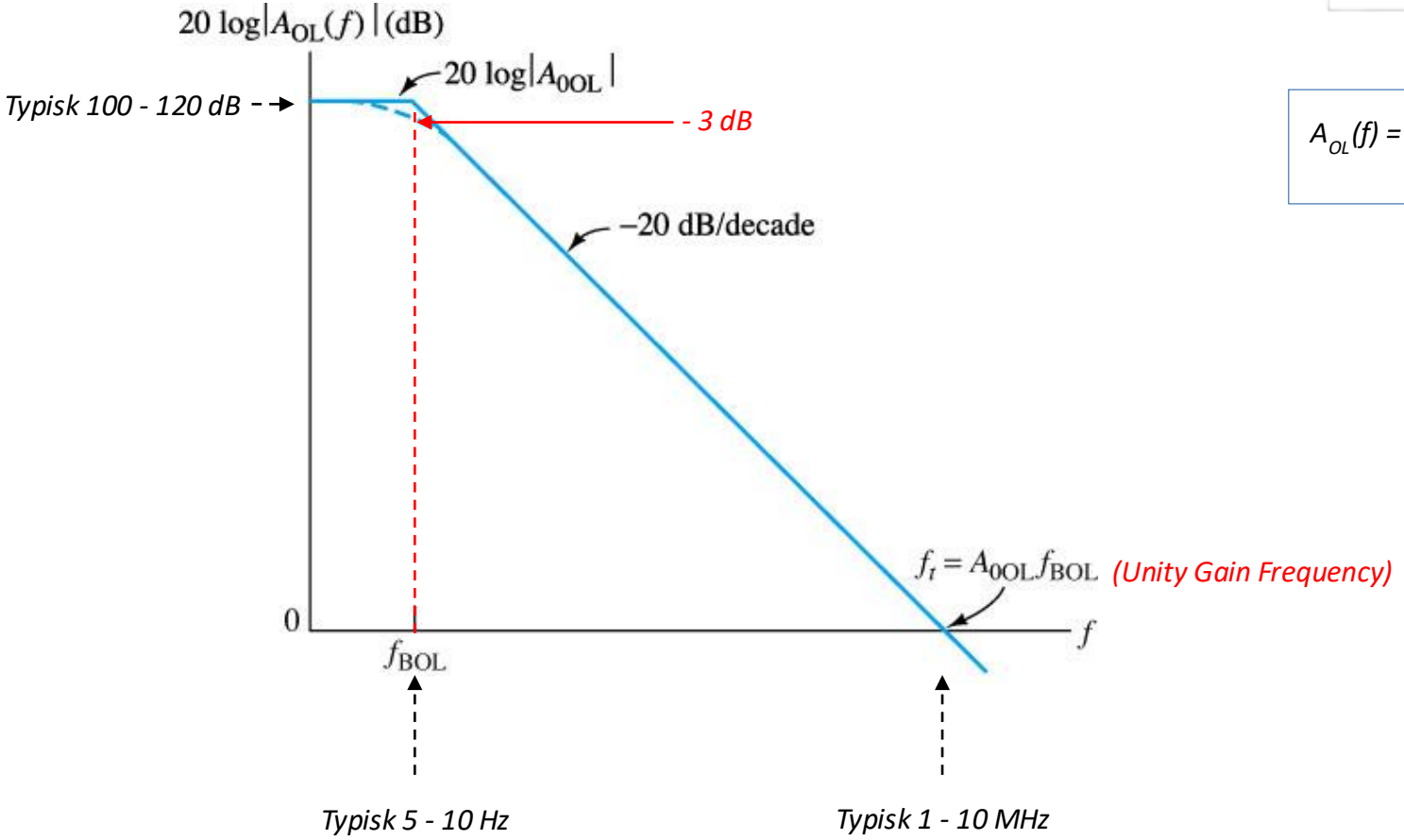
A_{CL} frekvensafhængig: $$A_{CL} = \frac{\alpha}{\beta} \cdot K_f$$



(Andre fejl på A_{OL} afspejler sig ligeledes i det samlede respons)

Åbensløjfeforstærkningen A_{OL}

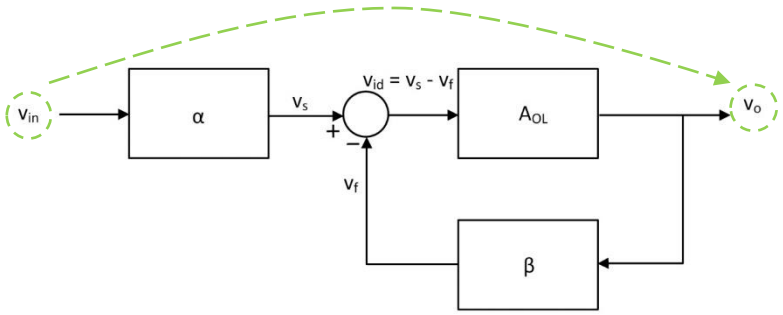


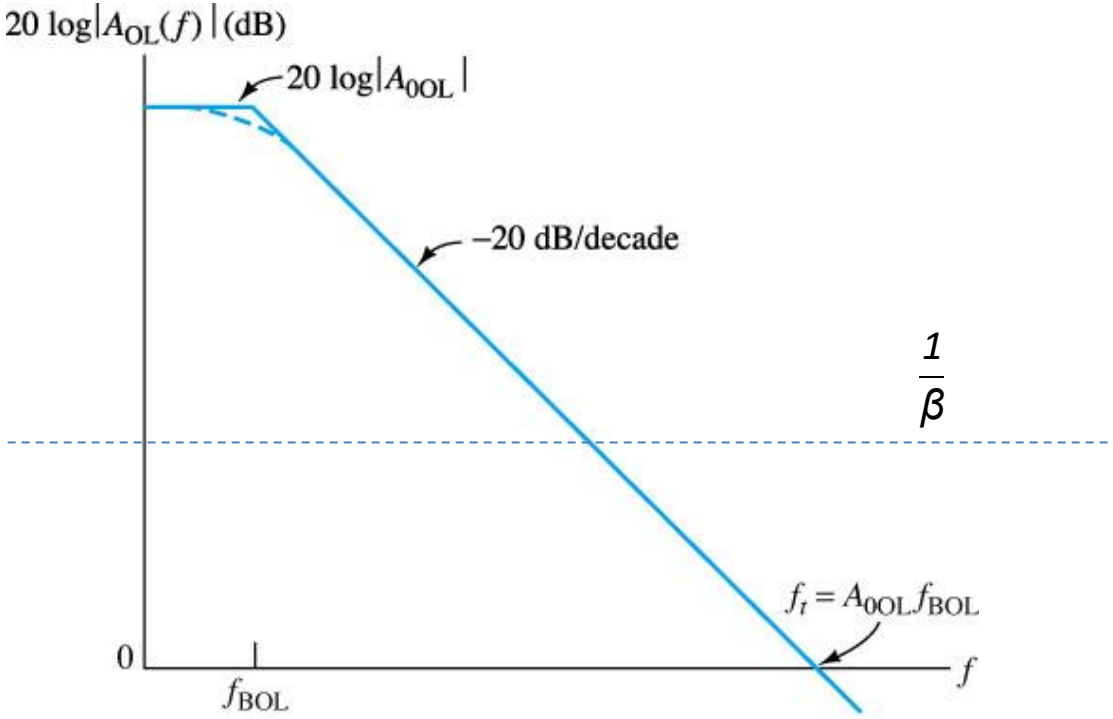


$$A_{OL}(f) = \frac{A_{0OL}}{1+j\frac{f}{f_{BOL}}}$$

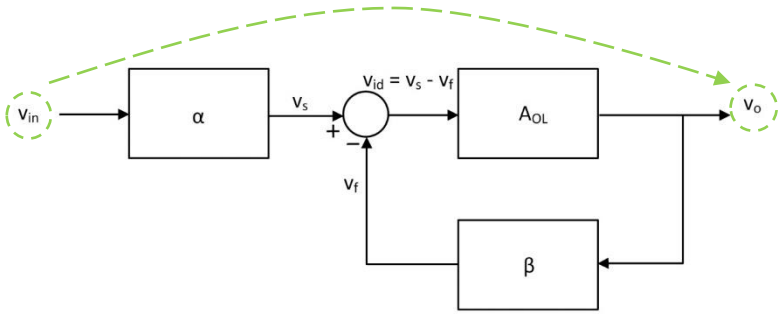
Konklusion: $f >> f_{BOL} \Rightarrow -90^{\circ}$

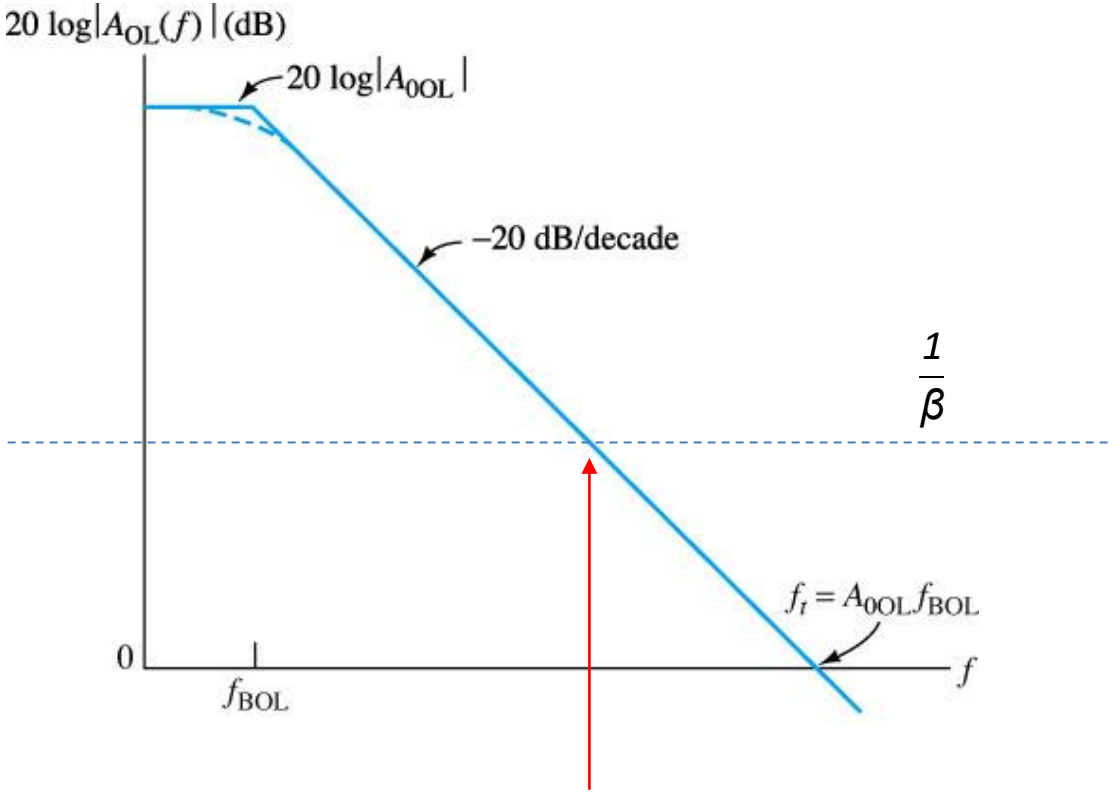
Lukketsløjfeforstærkningen A_{CL}





Lukketsløjfeforstærkningen A_{CL}

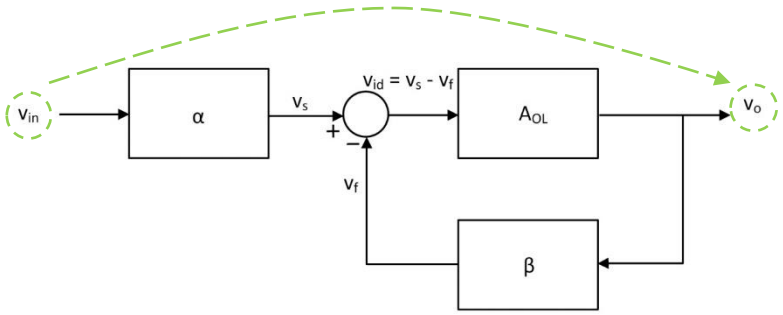


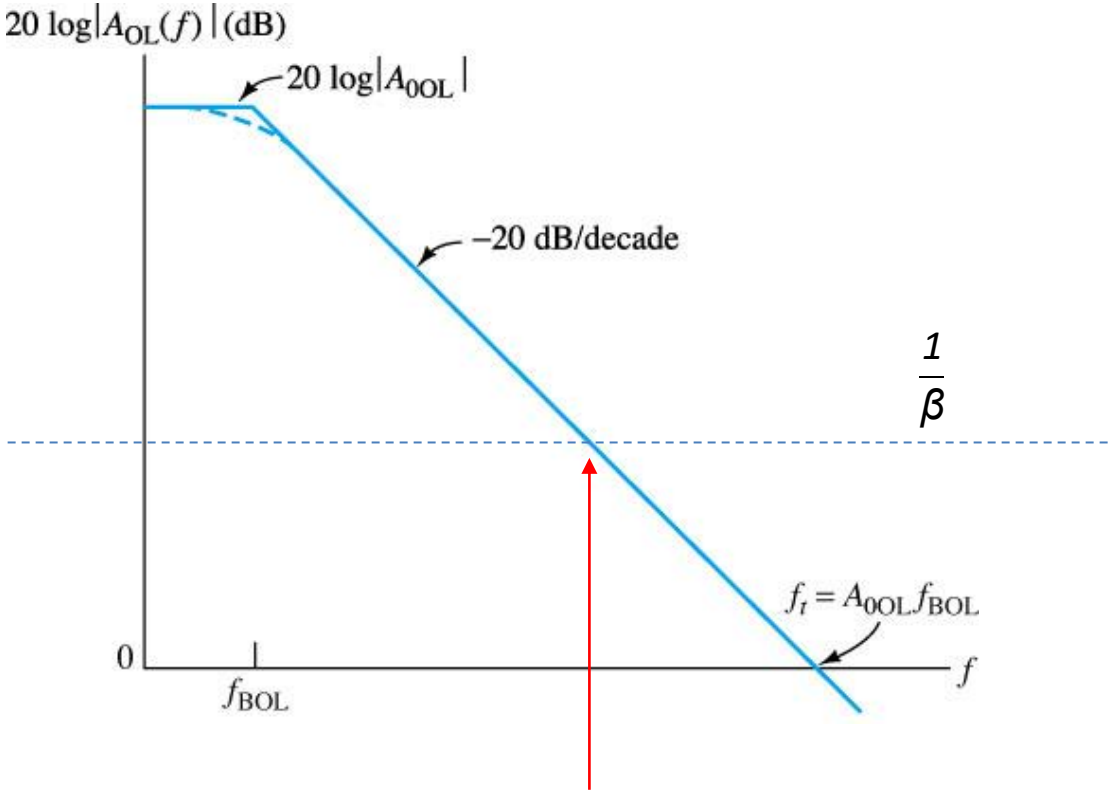


$$f_{BCL} = \beta \cdot f_t = \beta \cdot A_{0OL} \cdot f_{BOL}$$

Dette er 3dB-båndbredden/f_{3dB} for lukketsløjfeforstærkningen

Lukketsløjfeforstærkningen A_{CL}



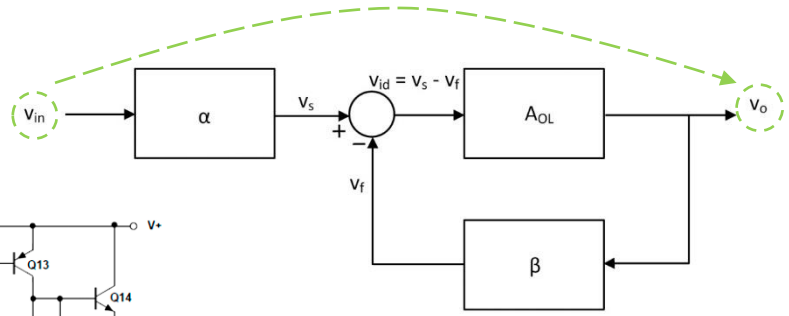


$$f_{BCL} = \beta \cdot f_t = \beta \cdot A_{0OL} \cdot f_{BOL}$$

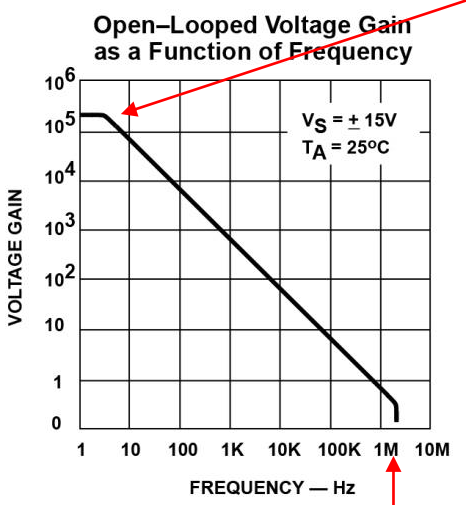
Dette er 3dB-båndbredden/f_{3dB} for lukketsløjfeforstærkningen

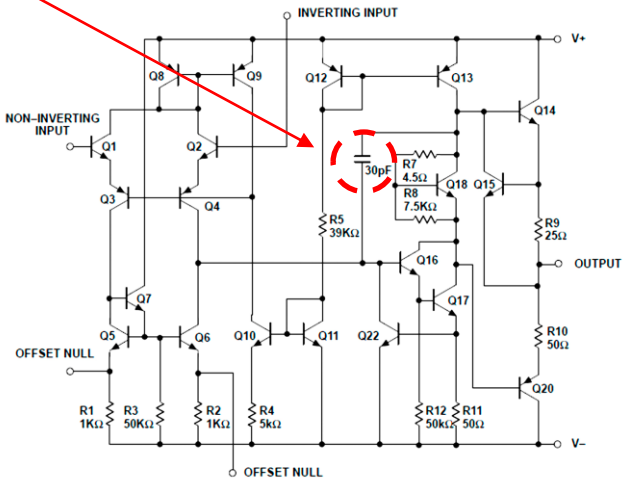
Hvorfor indgår α ikke i udtrykket for f_{BCL}?

Lukketsløjfeforstærkningen A_{CL}



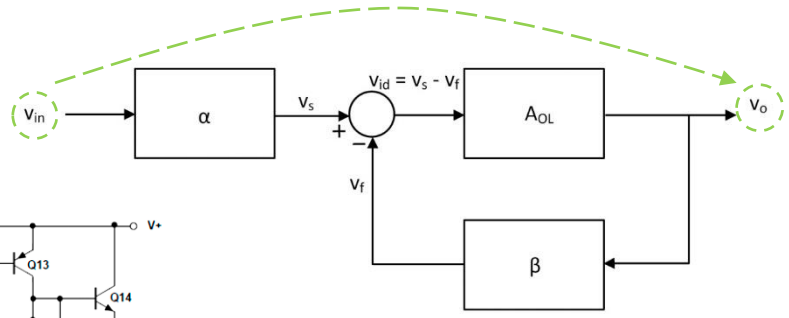
C_f

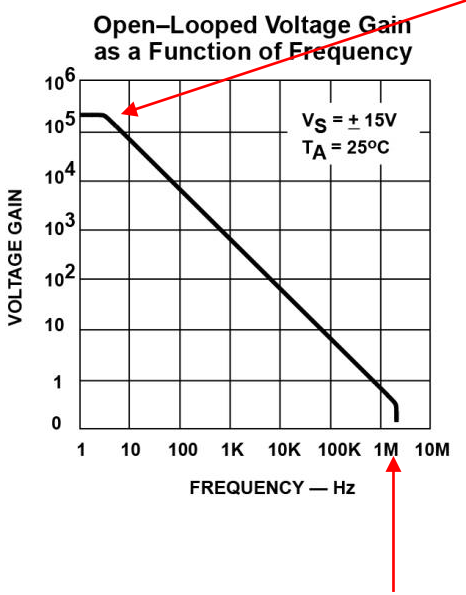




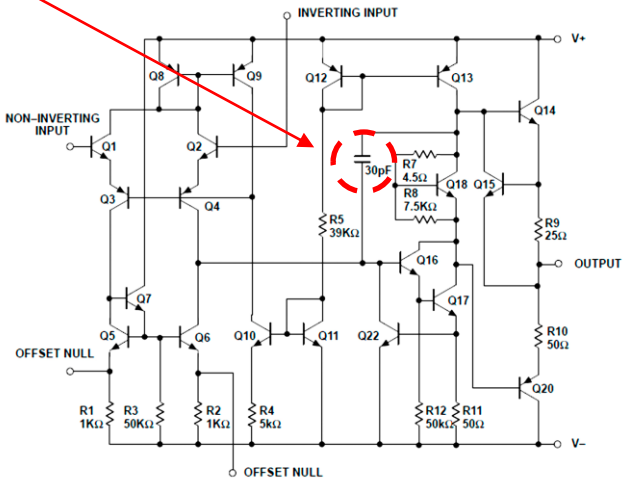
Interne kapaciteter (parasitiske)

Lukketsløjfeforstærkningen A_{CL}





C_f



Interne kapaciteter (parasitiske)

C_f kan evt. placeres eksternt for større båndbredde ved højere forstærkninger

$C_f = C_1$ ("Unity Gain" stabil) - databladsoplysning

Maksimal båndbredde ved $\frac{1}{\beta} > 1$: $C_f = \beta \cdot C_1$ ← *Dette giver samme båndbredde som for $C_f = C_1$ ved $\frac{1}{\beta} = 1$*

DC Bias

I praksis skal alle (aktive) kredsløb have en forsyningsspænding.

DC Bias

I praksis skal alle (aktive) kredsløb have en forsyningsspænding.



DC offset og muligvis drift

DC Bias

I praksis skal alle (aktive) kredsløb have en forsyningsspænding.



DC offset og muligvis drift

I sig selv en fejl hvis signalet man måler/regulerer på er lavfrekvent.

Eks. temperatur

DC Bias

I praksis skal alle (aktive) kredsløb have en forsyningsspænding.

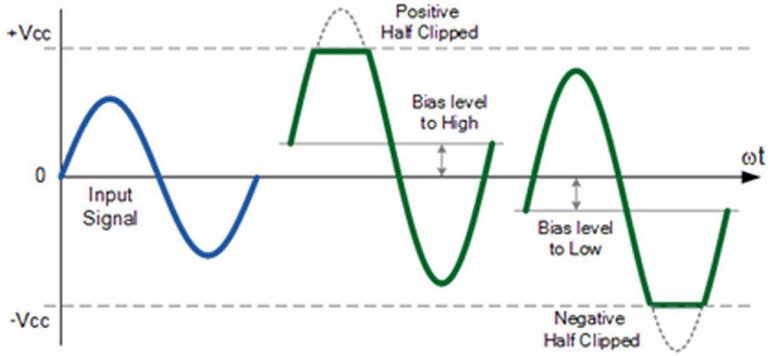
↓

DC offset og muligvis drift

I sig selv en fejl hvis signalet man måler/regulerer på er lavfrekvent.

Eks. temperatur

Indflydelse på signalbehandling i hele frekvensområdet:



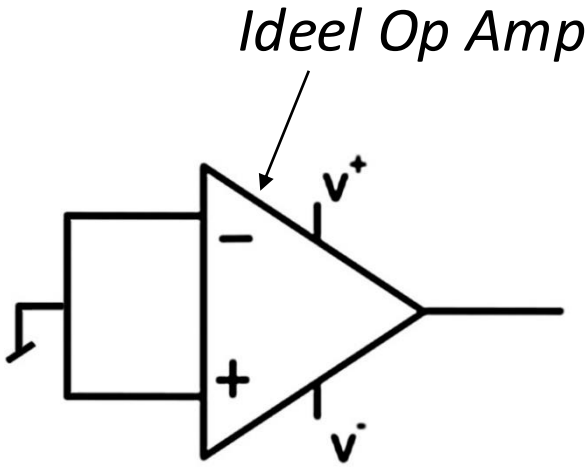
Offset- og biasfejl

Offsetspænding V_{off}
Biasstrøm I_B
Offsetstrøm I_{off}

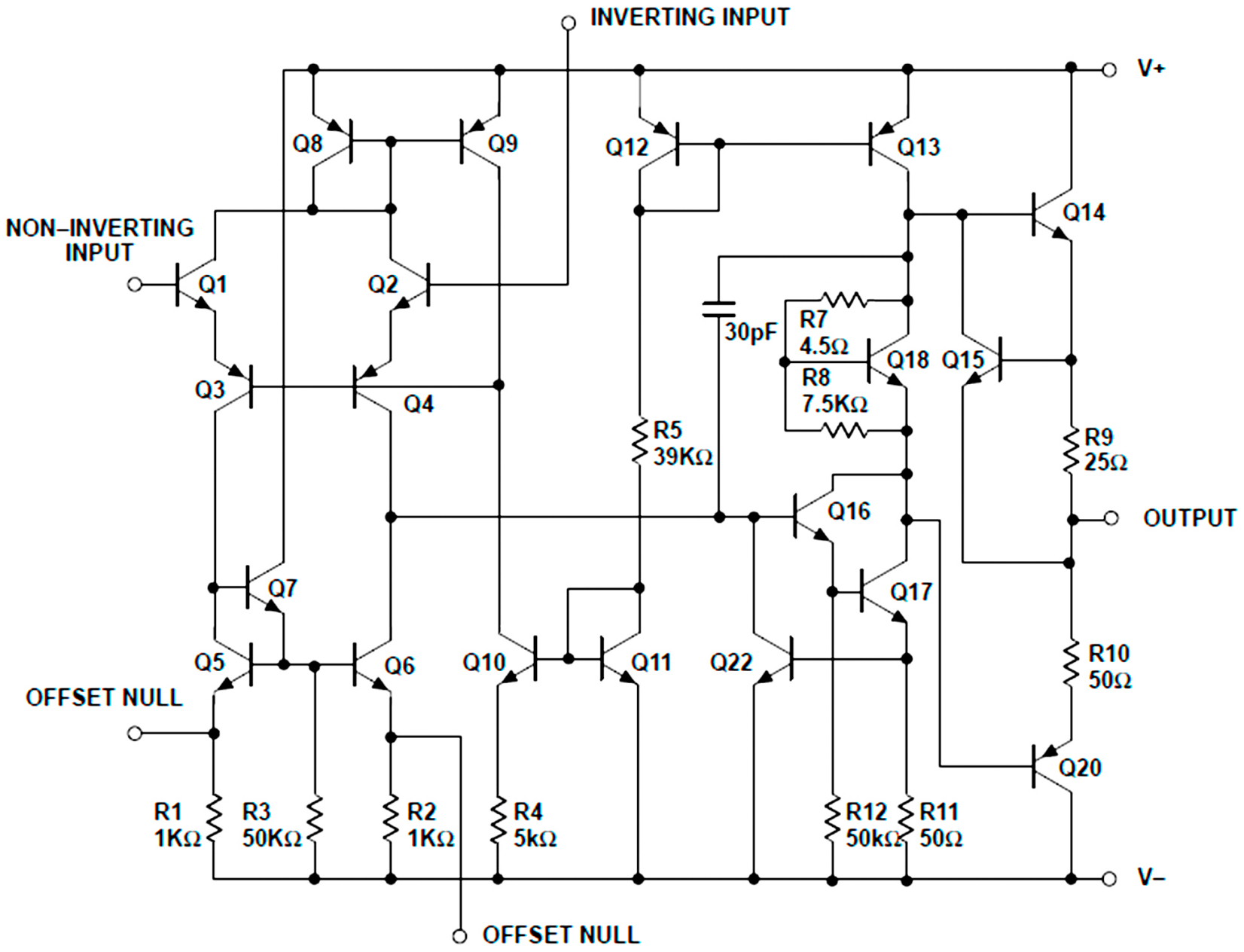


Betegnes ofte også V_{os} eller V_{io} i databladet

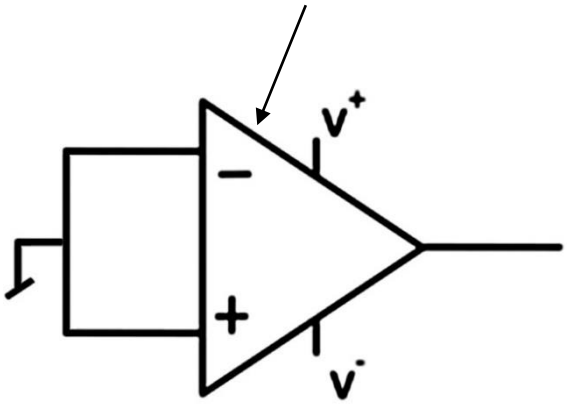
Kortslutning af de to indgangsterminaler

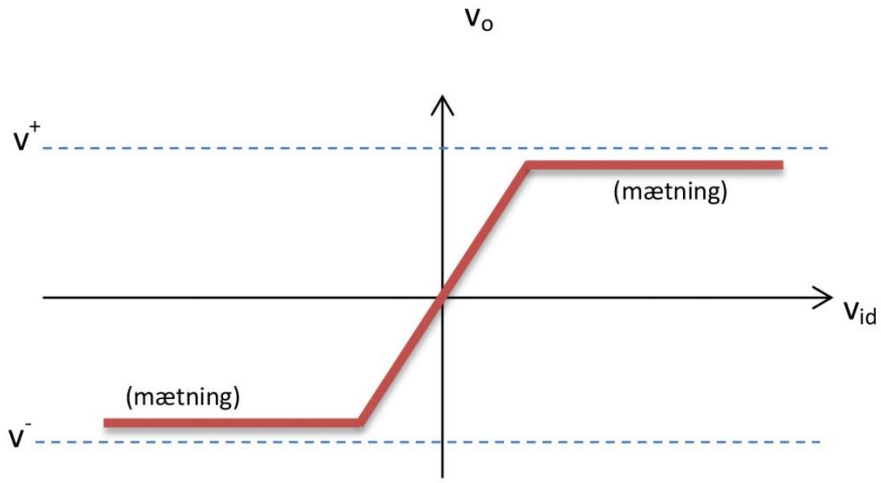


$v_{o\ ideel} = 0\ V$


INVERTING INPUT
V+
NON–INVERTING INPUT
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Q9
Q10
Q11
Q12
Q13
Q14
Q15
Q16
Q17
Q18
Q20
Q22
30pF
R7 4.5Ω
R8 7.5KΩ
R5 39KΩ
R9 25Ω
R10 50Ω
R1 1KΩ
R3 50KΩ
R2 1KΩ
R4 5kΩ
R12 50kΩ
R11 50Ω
OUTPUT
OFFSET NULL
OFFSET NULL
V–

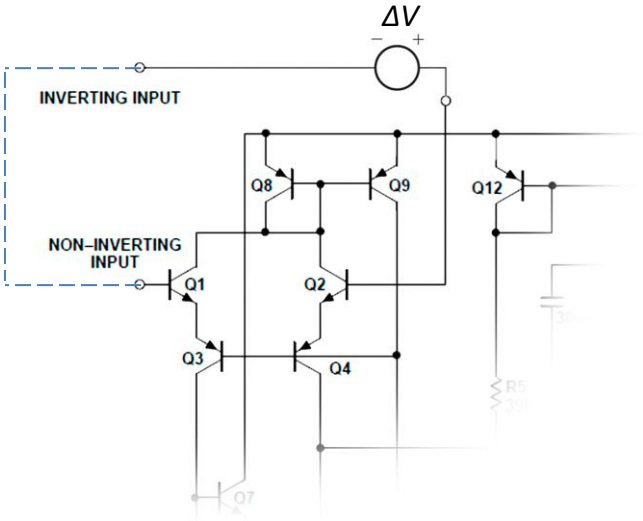
Praktisk Op Amp



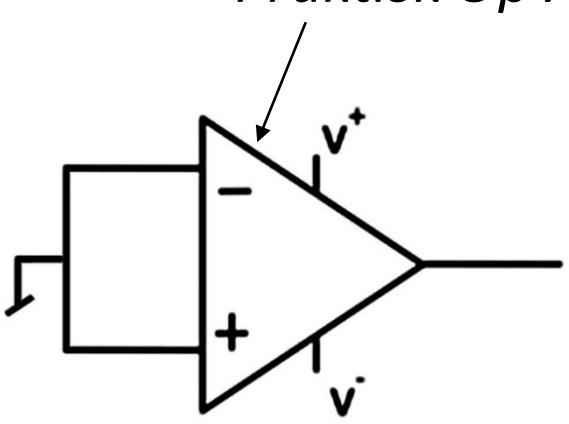


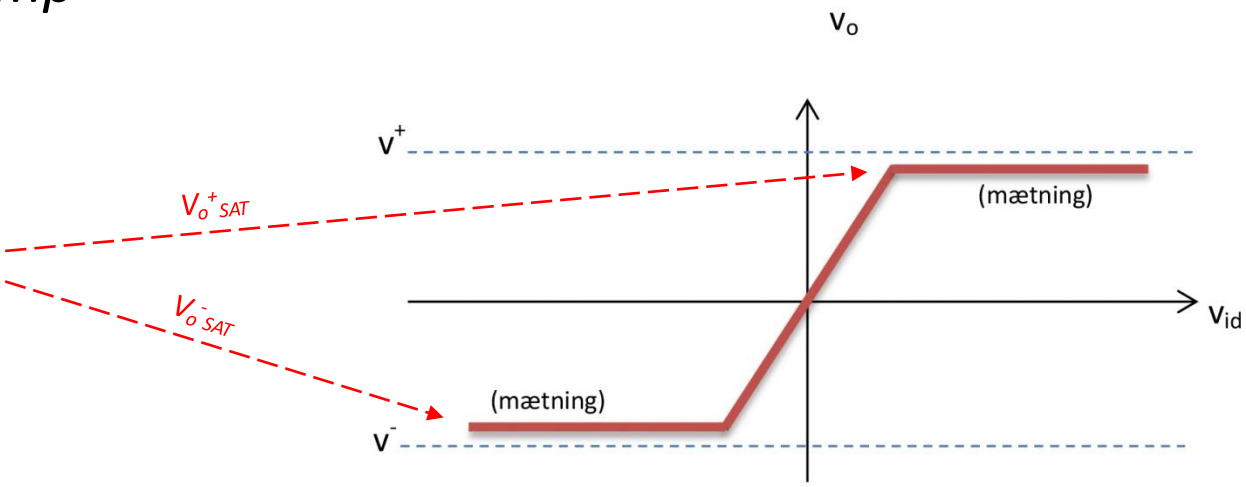
Mismatch ΔV i indgangstrinet

Meget stor A_{OL} (uden modkobling)



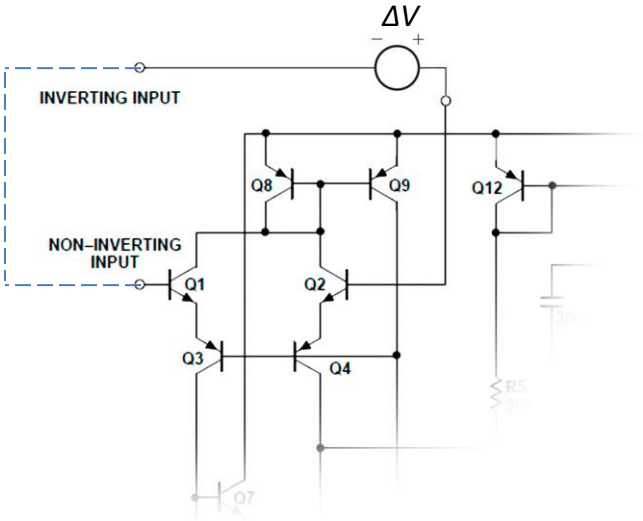
Praktisk Op Amp





Mismatch ΔV i indgangstrinet

Meget stor A_{OL} (uden modkobling)



Offsetspændingen V_{off}

Hvad er V_{off} for en størrelse – hvordan skal man tolke denne?

Datablad →

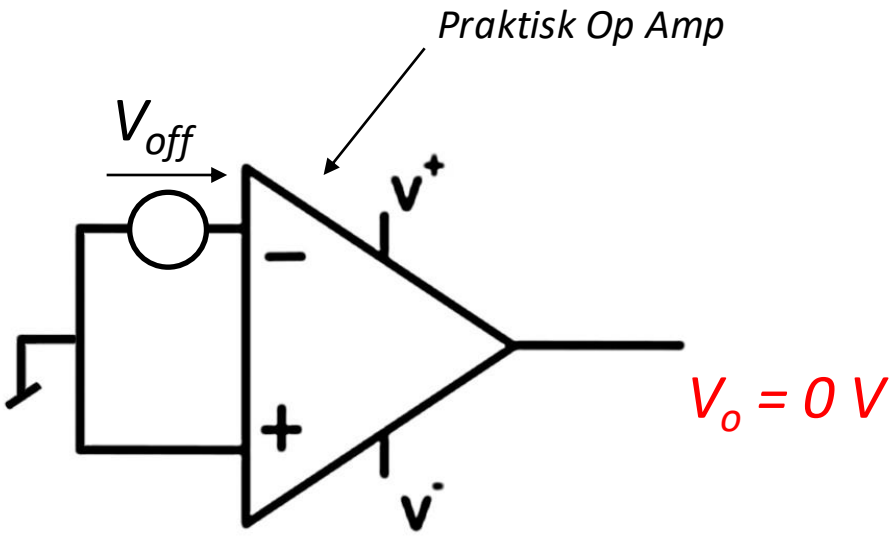
DC ELECTRICAL CHARACTERISTICS

T_A = 25°C, V_S = ±15V, unless otherwise specified.

SYMBOL	PARAMETER	TEST CONDITIONS	μA741			μA741C			UNIT
			Min	Typ	Max	Min	Typ	Max	
V_{OS}	Offset voltage	R_S=10kΩ		1.0	5.0		2.0	6.0	mV
		R_S=10kΩ, over temp.		1.0	6.0			7.5	mV
$\Delta V_{OS}/\Delta T$				10			10		μV/°C
I_{OS}	Offset current			20	200		20	200	nA
		Over temp.						300	nA
		T_A=+125°C		7.0	200				nA
		T_A=-55°C		20	500				nA
$\Delta I_{OS}/\Delta T$				200			200		pA/°C
I_{BIAS}	Input bias current			80	500		80	500	nA
		Over temp.						800	nA
		T_A=+125°C		30	500				nA
		T_A=-55°C		300	1500				nA
$\Delta I_B/\Delta T$				1			1		nA/°C
V_{OUT}	Output voltage swing	R_L=10kΩ	±12	±14		±12	±14		V
		R_L=2kΩ, over temp.	±10	±13		±10	±13		V
A_{VOL}	Large-signal voltage gain	R_L=2kΩ, V_O=±10V	50	200		20	200		V/mV
		R_L=2kΩ, V_O=±10V, over temp.	25			15			V/mV
	Offset voltage adjustment range			±30			±30		mV
PSRR	Supply voltage rejection ratio	R_S≤10kΩ					10	150	μV/V
		R_S≤10kΩ, over temp.		10	150				μV/V
CMRR	Common-mode rejection ratio					70	90		dB
		Over temp.	70	90					dB
I_{CC}	Supply current			1.4	2.8		1.4	2.8	mA
		T_A=+125°C		1.5	2.5				mA
		T_A=-55°C		2.0	3.3				mA
V_{IN}	Input voltage range	(μA741, over temp.)	±12	±13		±12	±13		V
R_{IN}	Input resistance		0.3	2.0		0.3	2.0		MΩ
P_D	Power consumption			50	85		50	85	mW
		T_A=+125°C		45	75				mW
		T_A=-55°C		45	100				mW
R_{OUT}	Output resistance			75			75		Ω
I_{SC}	Output short-circuit current		10	25	60	10	25	60	mA

Offsetspændingen V_{off}

Hvad er V_{off} for en størrelse – hvordan skal man tolke denne?



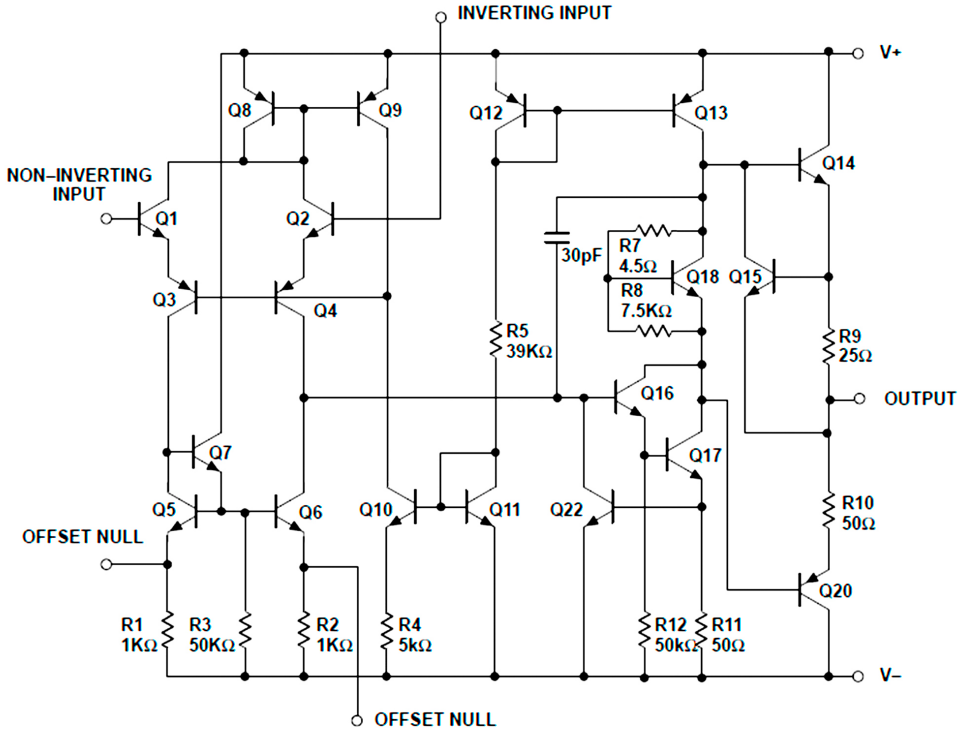
V_{off} "ophæver" Op Amp'ens egen indgangsoffsetspænding

Dual *henholdsvis* ***singulær*** *størrelse?*

Dual: *Kan være* ***enten*** *positiv eller negativ*

Singulær: *Kan* ***ikke*** *skifte fortegn*

V_{off} **:** ***Dual***



DC ELECTRICAL CHARACTERISTICS

T_A = 25°C, V_S = ±15V, unless otherwise specified.

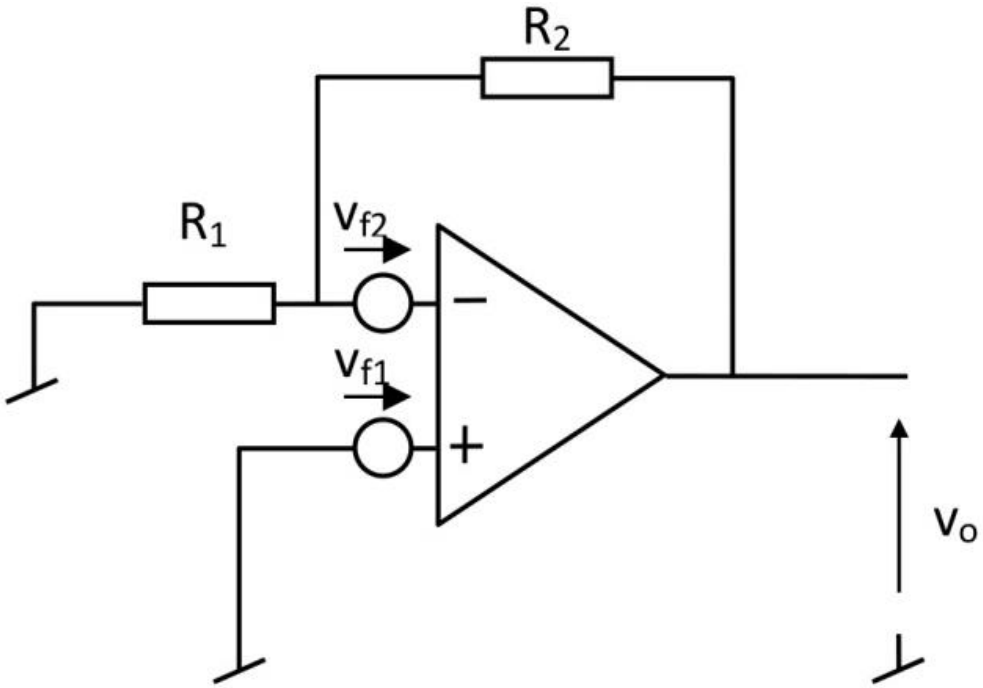
SYMBOL	PARAMETER	TEST CONDITIONS	μA741			μA741C			UNIT
			Min	Typ	Max	Min	Typ	Max	
V_{OS}	Offset voltage	R_S=10kΩ		1.0	5.0		2.0	6.0	mV
		R_S=10kΩ, over temp.		1.0	6.0			7.5	mV
$\Delta V_{OS}/\Delta T$				10			10		μV/°C
I_{OS}	Offset current			20	200		20	200	nA
		Over temp.						300	nA
		T_A=+125°C		7.0	200				nA
		T_A=-55°C		20	500				nA
$\Delta I_{OS}/\Delta T$				200			200		pA/°C
I_{BIAS}	Input bias current			80	500		80	500	nA
		Over temp.						800	nA
		T_A=+125°C		30	500				nA
		T_A=-55°C		300	1500				nA
$\Delta I_B/\Delta T$				1			1		nA/°C
V_{OUT}	Output voltage swing	R_L=10kΩ	±12	±14		±12	±14		V

Typisk værdi for V_{off} for en μA741C: *± 2,0 mV*

Max værdi for V_{off} for en μA741C: *± 6,0 mV*

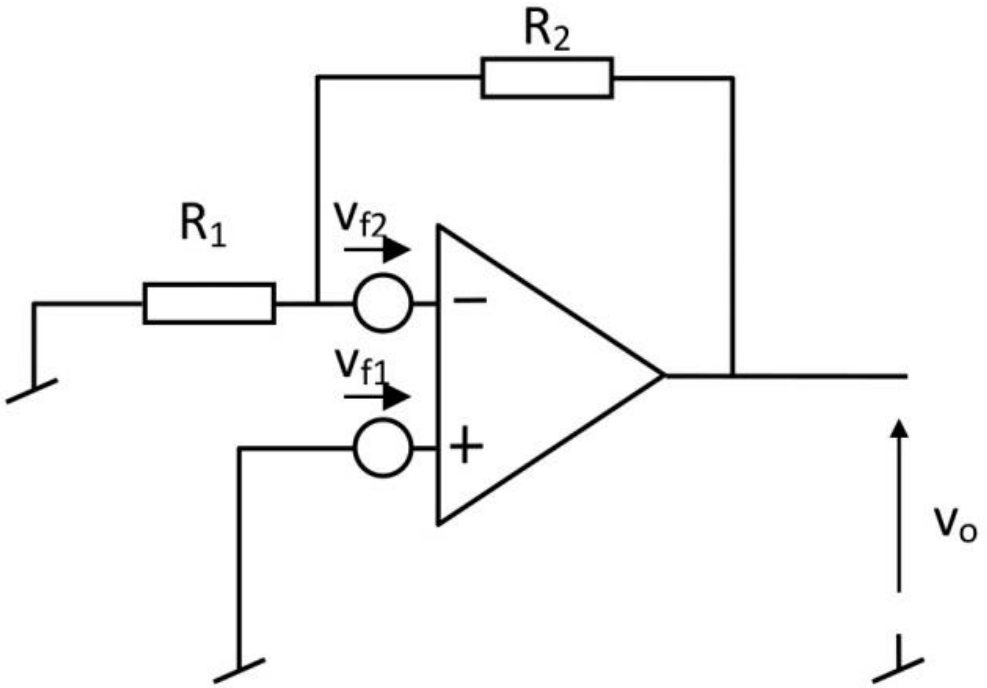
Sp. 5 fra forberedelsen:

Betragt følgende ideelle Op Amp med de to påtrykte spændinger V_{f1} og V_{f2}



Sp. 5 fra forberedelsen:

Betragt følgende ideelle Op Amp med de to påtrykte spændinger V_{f1} og V_{f2}

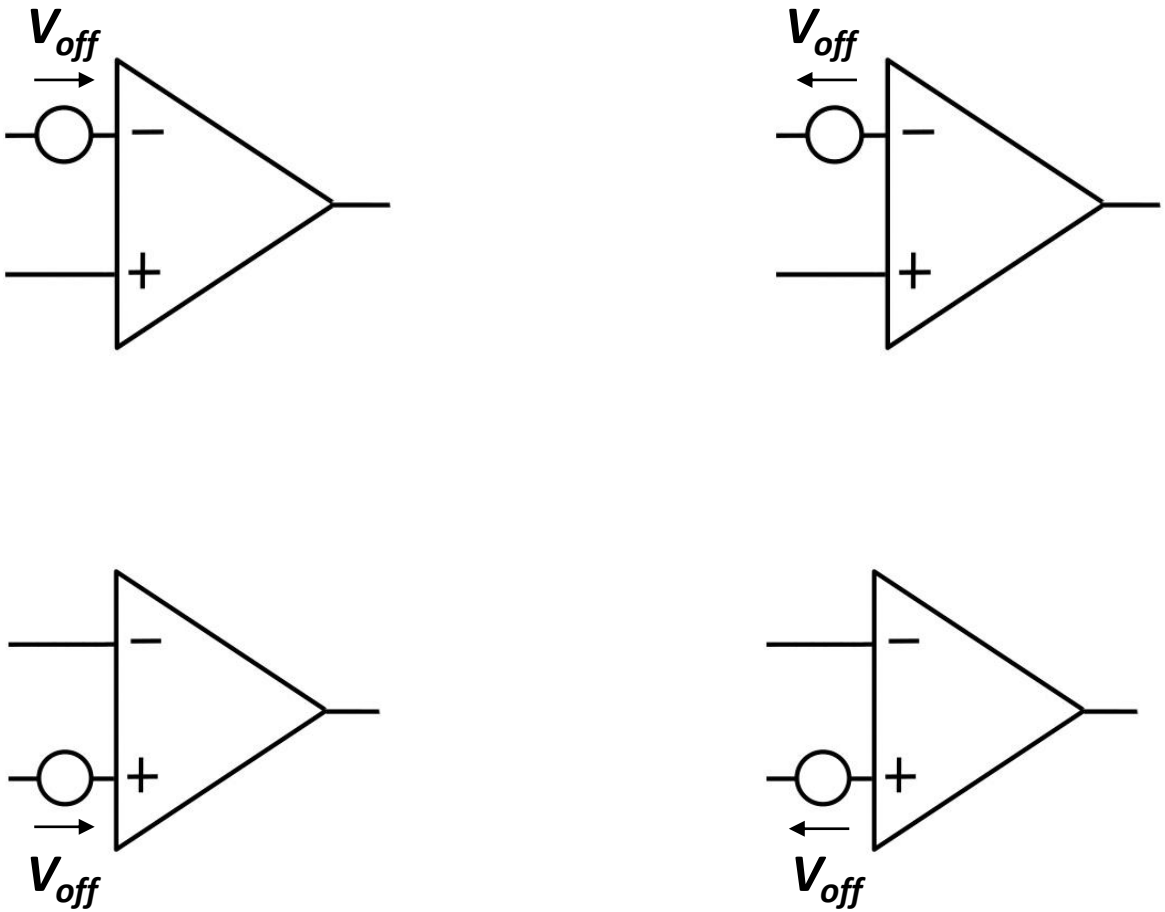


$$v_o = \frac{\alpha_1}{\beta} \cdot V_{f1} + \frac{\alpha_2}{\beta} \cdot V_{f2}$$

Feedback system på tavle →

Find α_1 og α_2 og vurdér herefter forstærkningen af de to spændinger til udgangen.

Placering af V_{off} i ækvivalent diagram



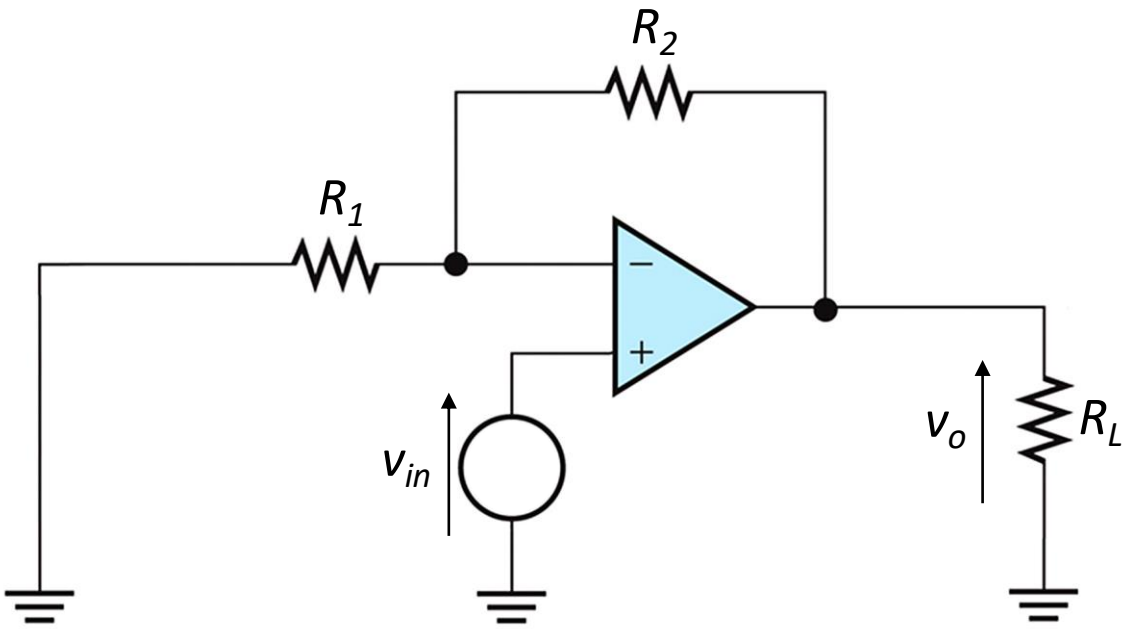
Eksempel:

$V_{off} = \pm 2\ mV$

$R_1 = 1\ k\Omega$

$R_2 = 100\ k\Omega$

$R_L = 2\ k\Omega$

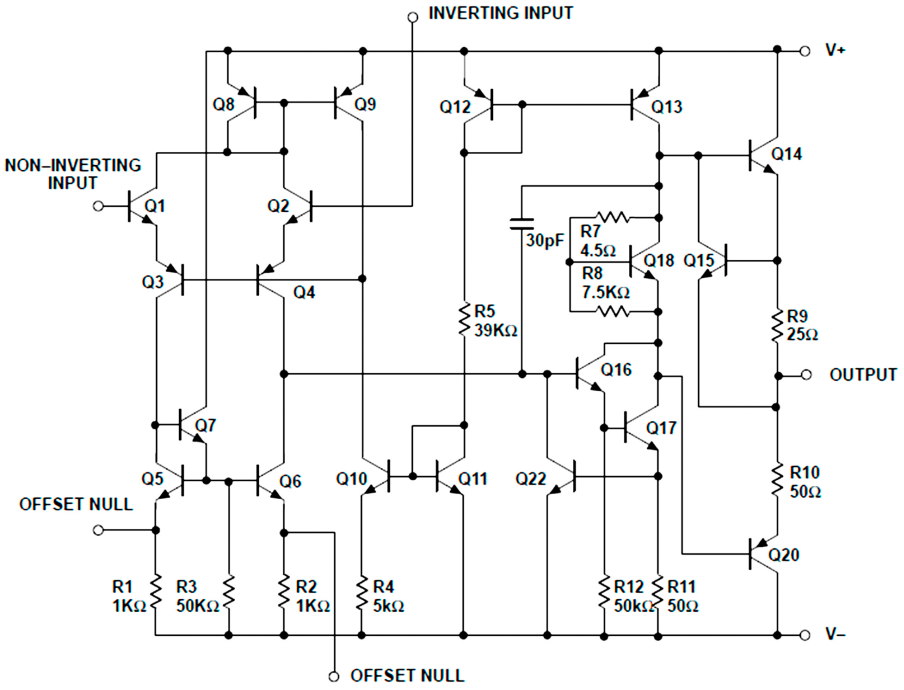


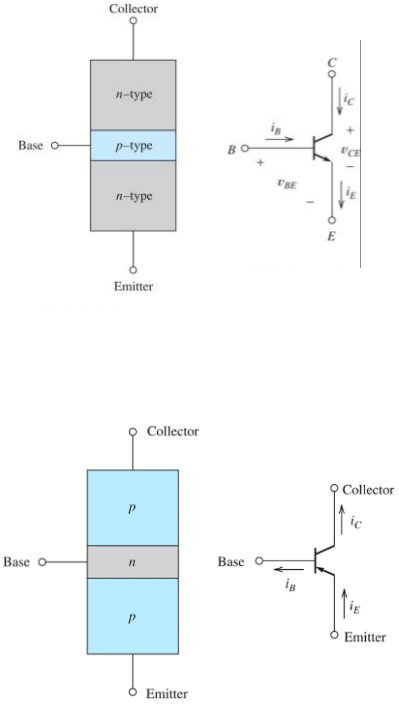
Beregn fejlbidrag på udgangen hidrørende fra V_{off}

Midtvejsdiskussion

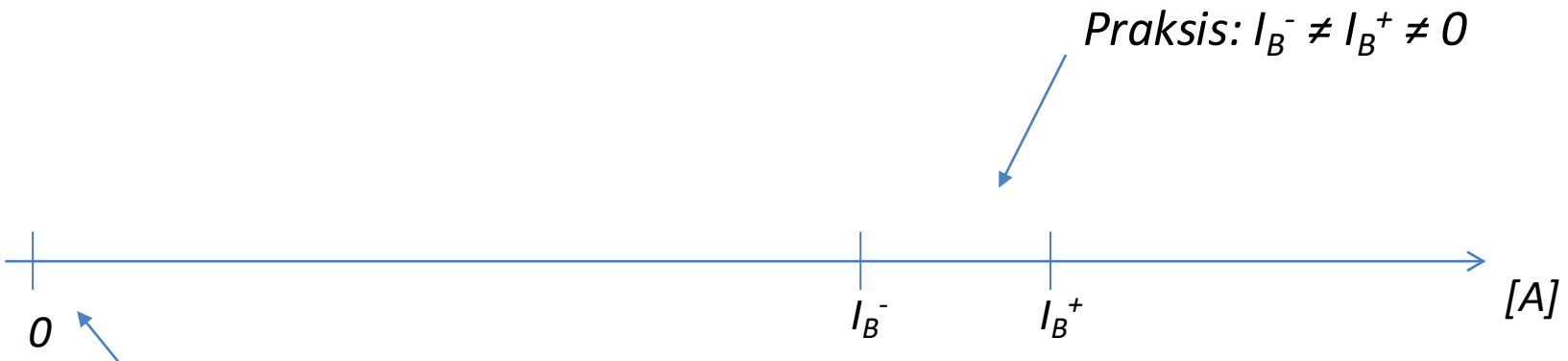
Biasstrøm I_B

*Strøm ind i **ideel** Op Amp: 0 A*

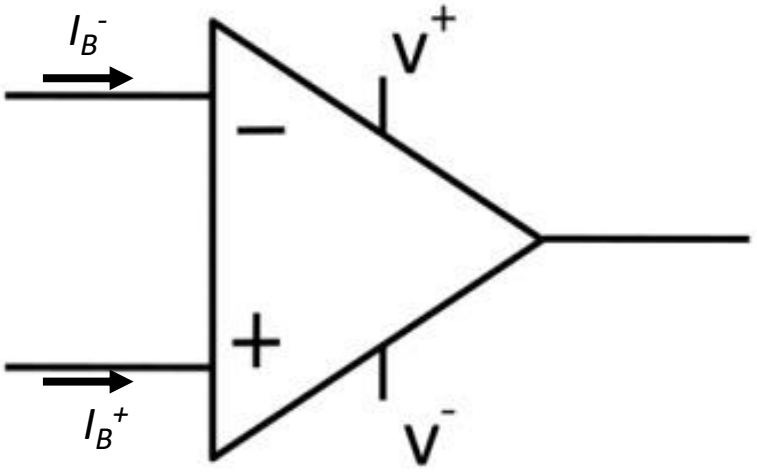




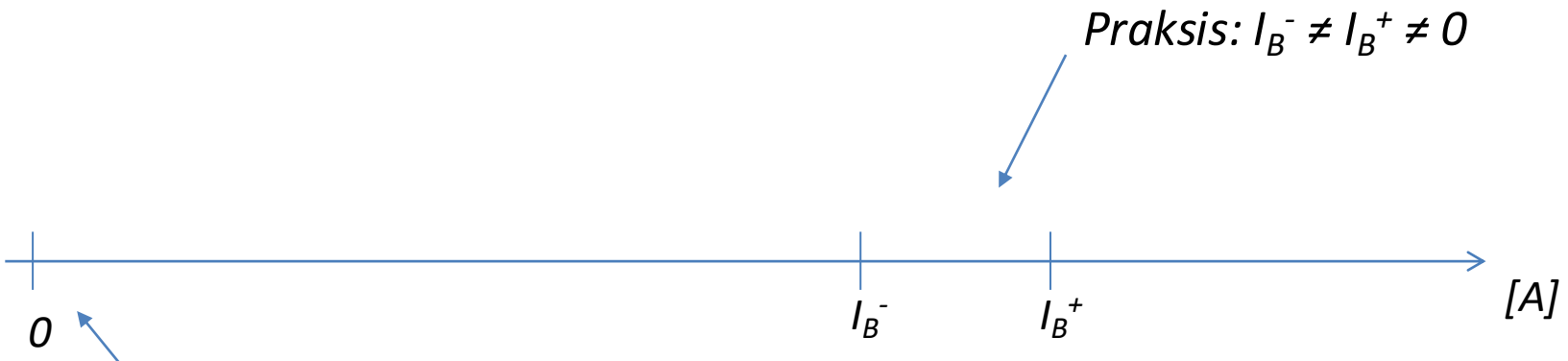
Biasstrømmene I_B^- og I_B^+



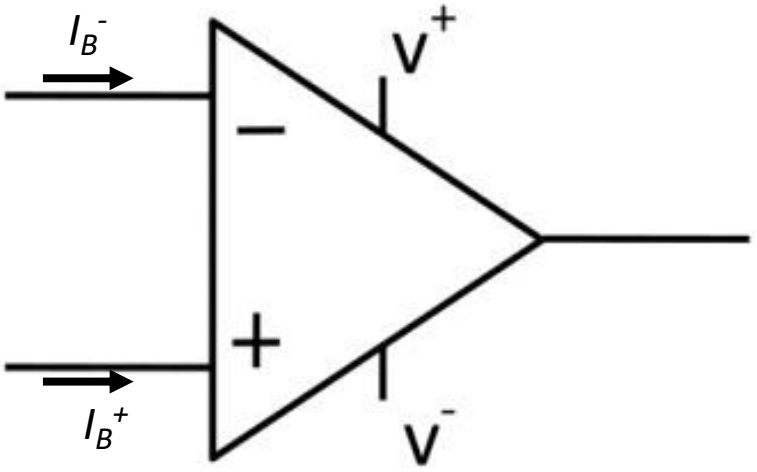
Ideelt: $I_B^- = I_B^+ = 0$



Biasstrømmene I_B^- og I_B^+

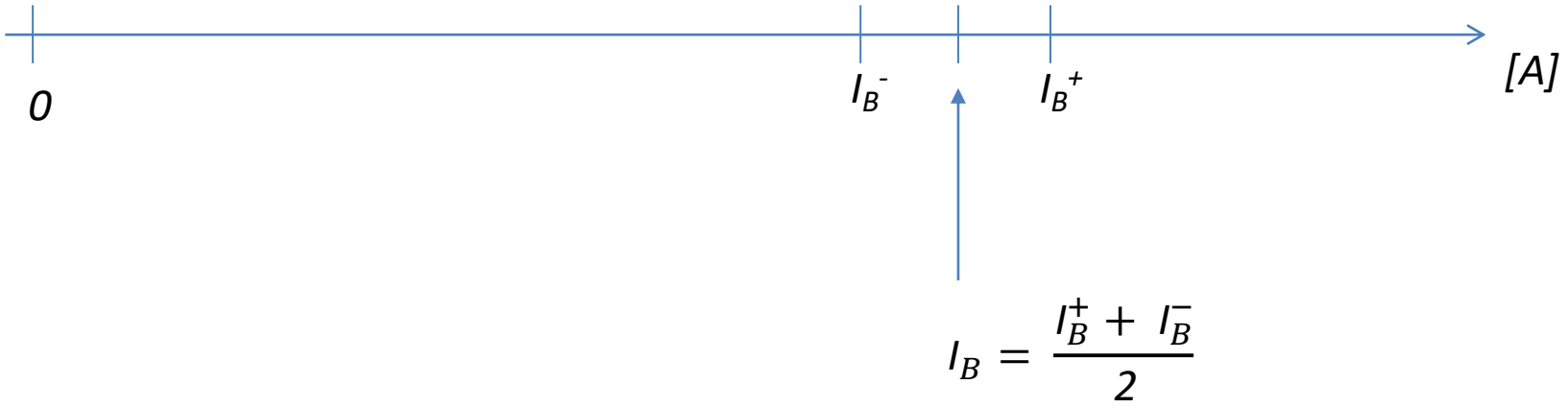


Ideelt: $I_B^- = I_B^+ = 0$

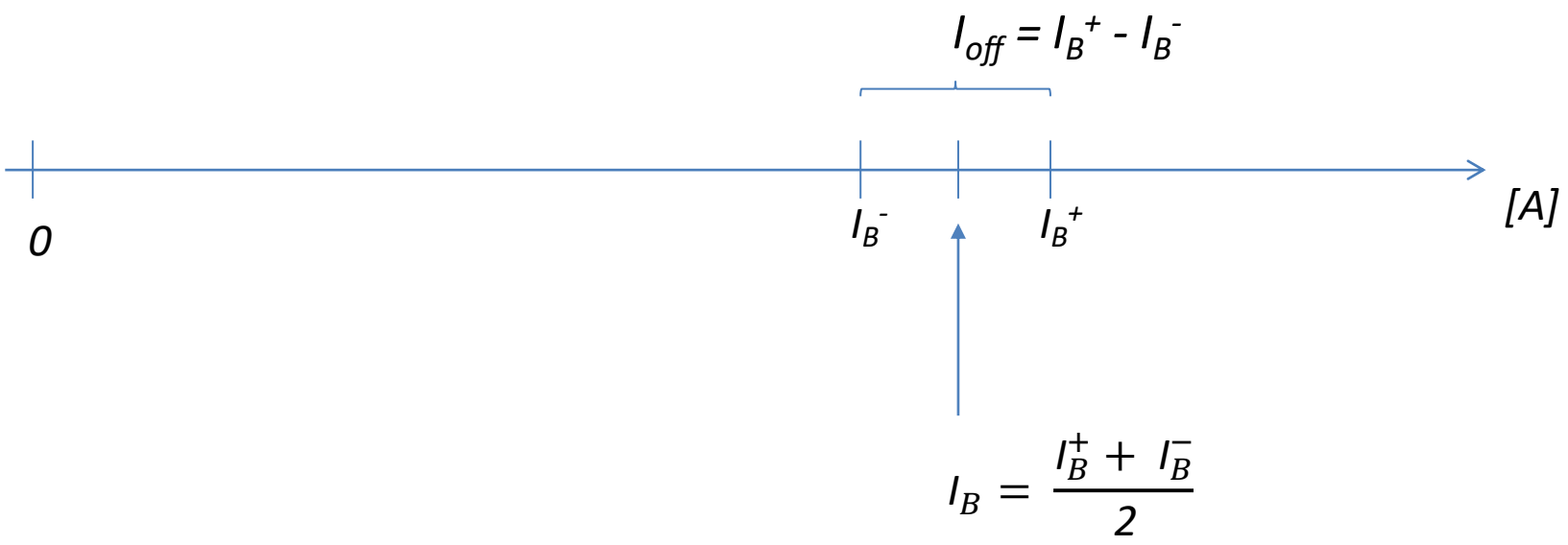


Man får IKKE oplyst I_B^- og I_B^+ i databladet!

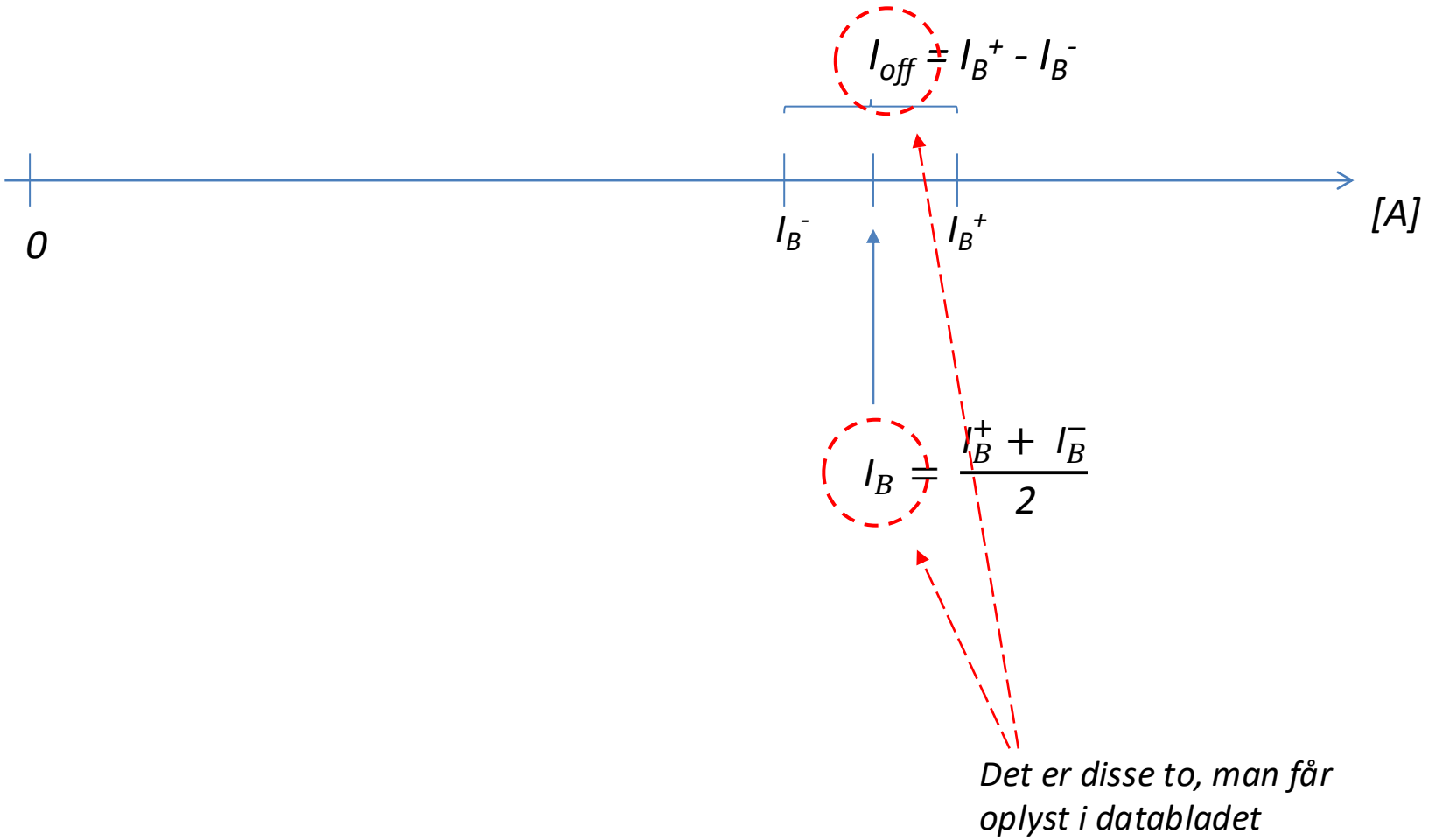
Biasstrømmene I_B^- og I_B^+



Biasstrømmene I_B^- og I_B^+



Biasstrømmene I_B^- og I_B^+

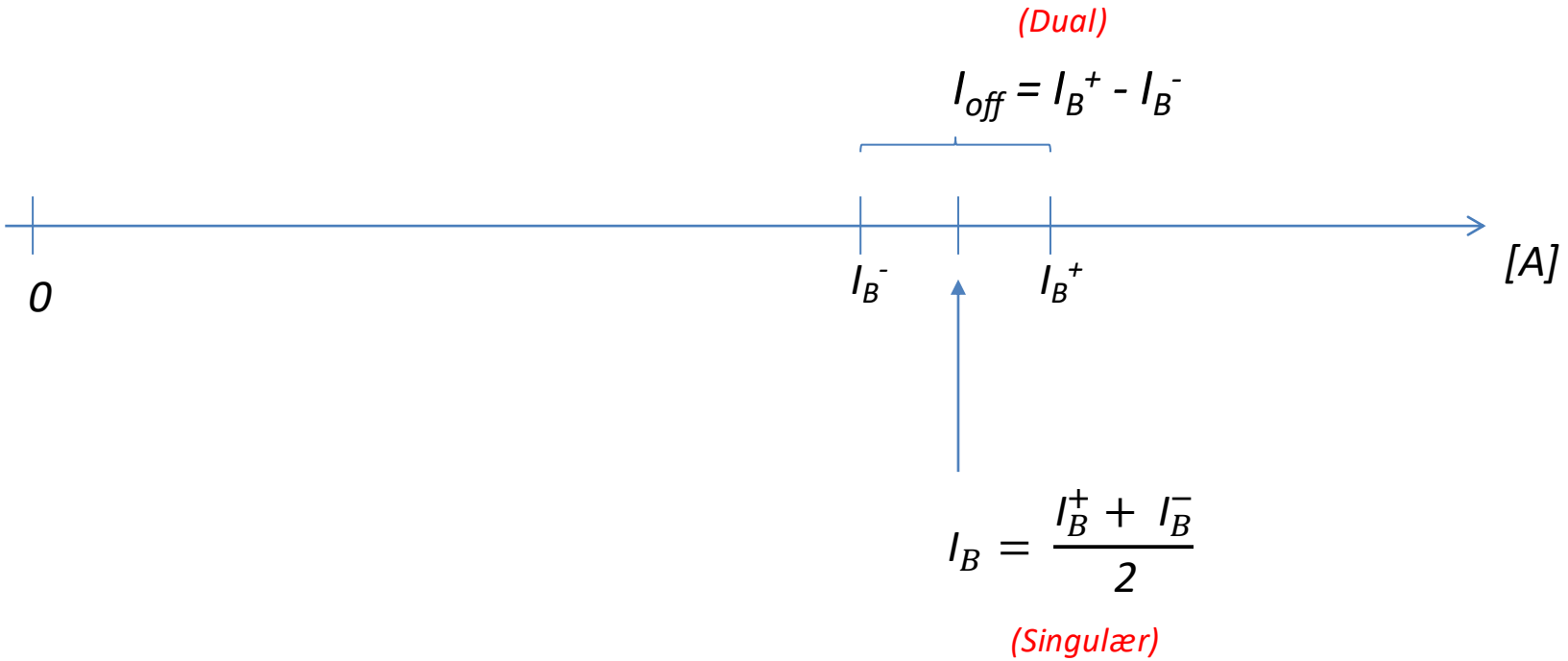


DC ELECTRICAL CHARACTERISTICS

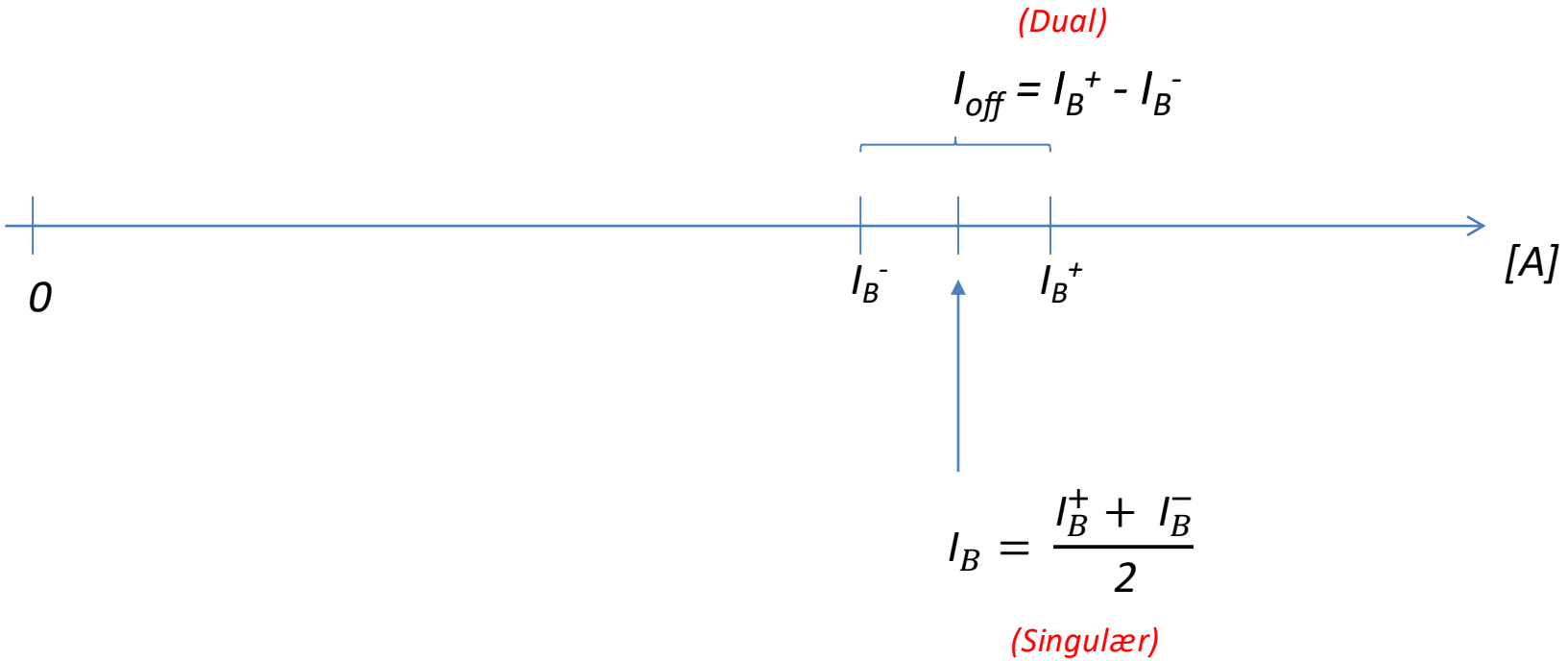
T_A = 25°C, V_S = ±15V, unless otherwise specified.

SYMBOL	PARAMETER	TEST CONDITIONS	μA741			μA741C			UNIT
			Min	Typ	Max	Min	Typ	Max	
V_{OS}	Offset voltage	R_S=10kΩ		1.0	5.0		2.0	6.0	mV
		R_S=10kΩ, over temp.		1.0	6.0			7.5	mV
$\Delta V_{OS}/\Delta T$				10			10		μV/°C
I_{OS}	Offset current			20	200		20	200	nA
		Over temp.						300	nA
		T_A=+125°C		7.0	200				nA
		T_A=-55°C		20	500				nA
$\Delta I_{OS}/\Delta T$				200			200		pA/°C
I_{BIAS}	Input bias current			80	500		80	500	nA
		Over temp.						800	nA
		T_A=+125°C		30	500				nA
		T_A=-55°C		300	1500				nA
$\Delta I_B/\Delta T$				1			1		nA/°C
V_{OUT}	Output voltage swing	R_L=10kΩ	±12	±14		±12	±14		V
		R_L=2kΩ, over temp.	±10	±13		±10	±13		V
A_{VOL}	Large-signal voltage gain	R_L=2kΩ, V_O=±10V	50	200		20	200		V/mV
		R_L=2kΩ, V_O=±10V, over temp.	25			15			V/mV
	Offset voltage adjustment range			±30			±30		mV
PSRR	Supply voltage rejection ratio	R_S≤10kΩ					10	150	μV/V
		R_S≤10kΩ, over temp.		10	150				μV/V
CMRR	Common-mode rejection ratio					70	90		dB
		Over temp.	70	90					dB
I_{CC}	Supply current			1.4	2.8		1.4	2.8	mA
		T_A=+125°C		1.5	2.5				mA
		T_A=-55°C		2.0	3.3				mA
V_{IN}	Input voltage range	(μA741, over temp.)	±12	±13		±12	±13		V
R_{IN}	Input resistance		0.3	2.0		0.3	2.0		MΩ
P_D	Power consumption			50	85		50	85	mW
		T_A=+125°C		45	75				mW
		T_A=-55°C		45	100				mW
R_{OUT}	Output resistance			75			75		Ω
I_{SC}	Output short-circuit current		10	25	60	10	25	60	mA

Biasstrømmene I_B^- og I_B^+



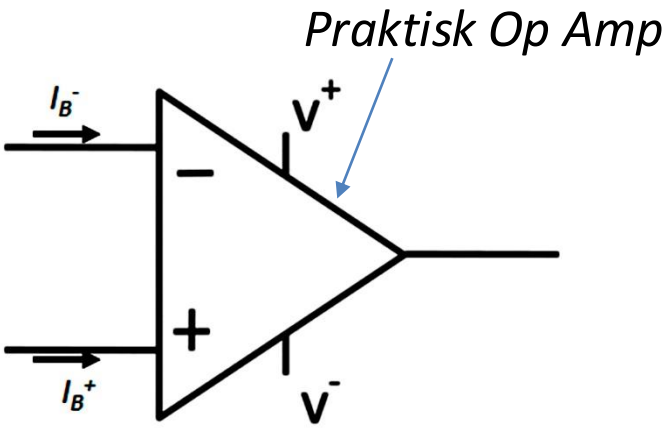
Biasstrømmene I_B^- og I_B^+



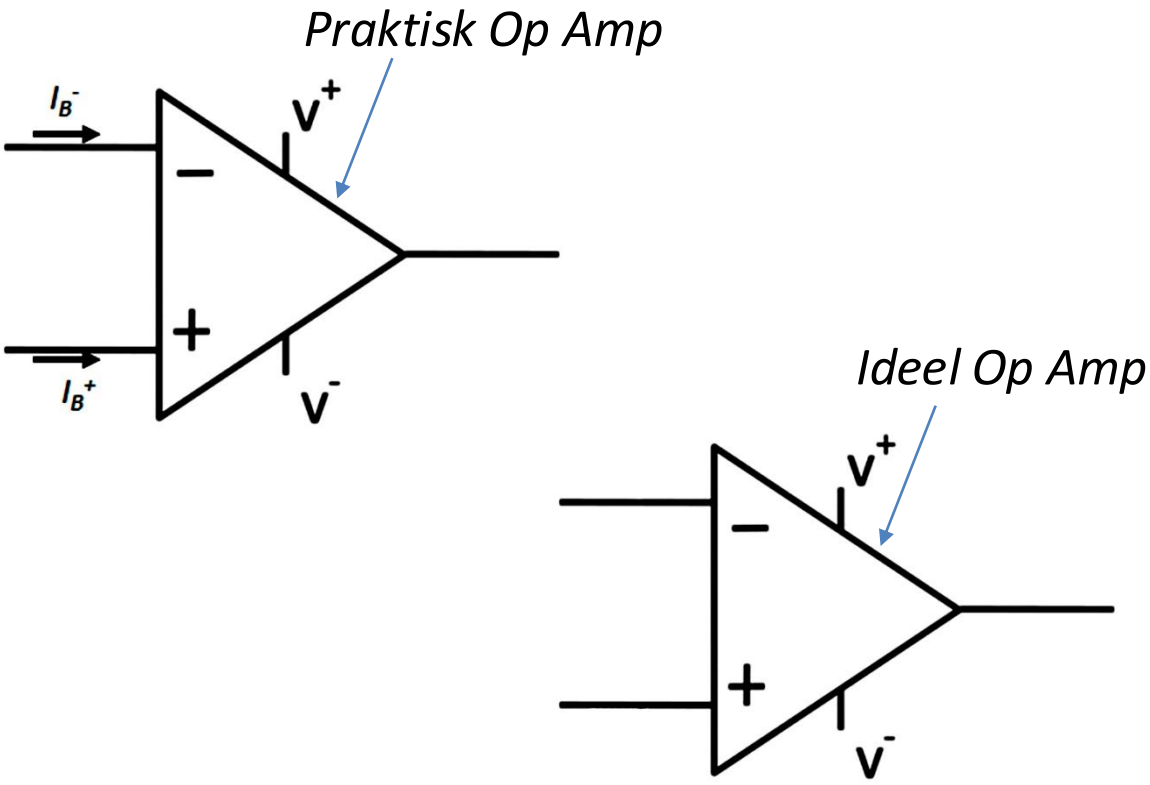
$$I_B^+ = I_B + \frac{I_{off}}{2}$$

$$I_B^- = I_B - \frac{I_{off}}{2}$$

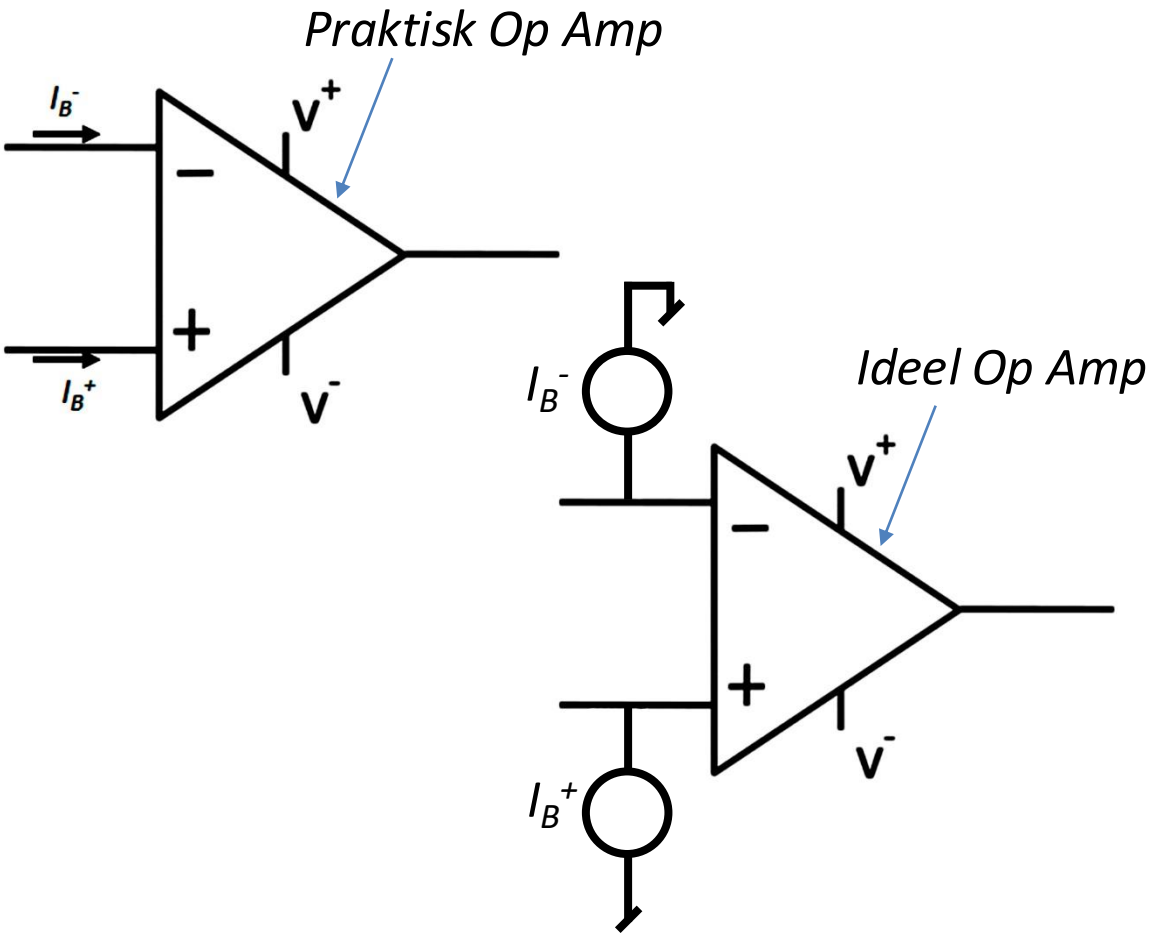
Placering af biasstrømme i fejlækvivalent



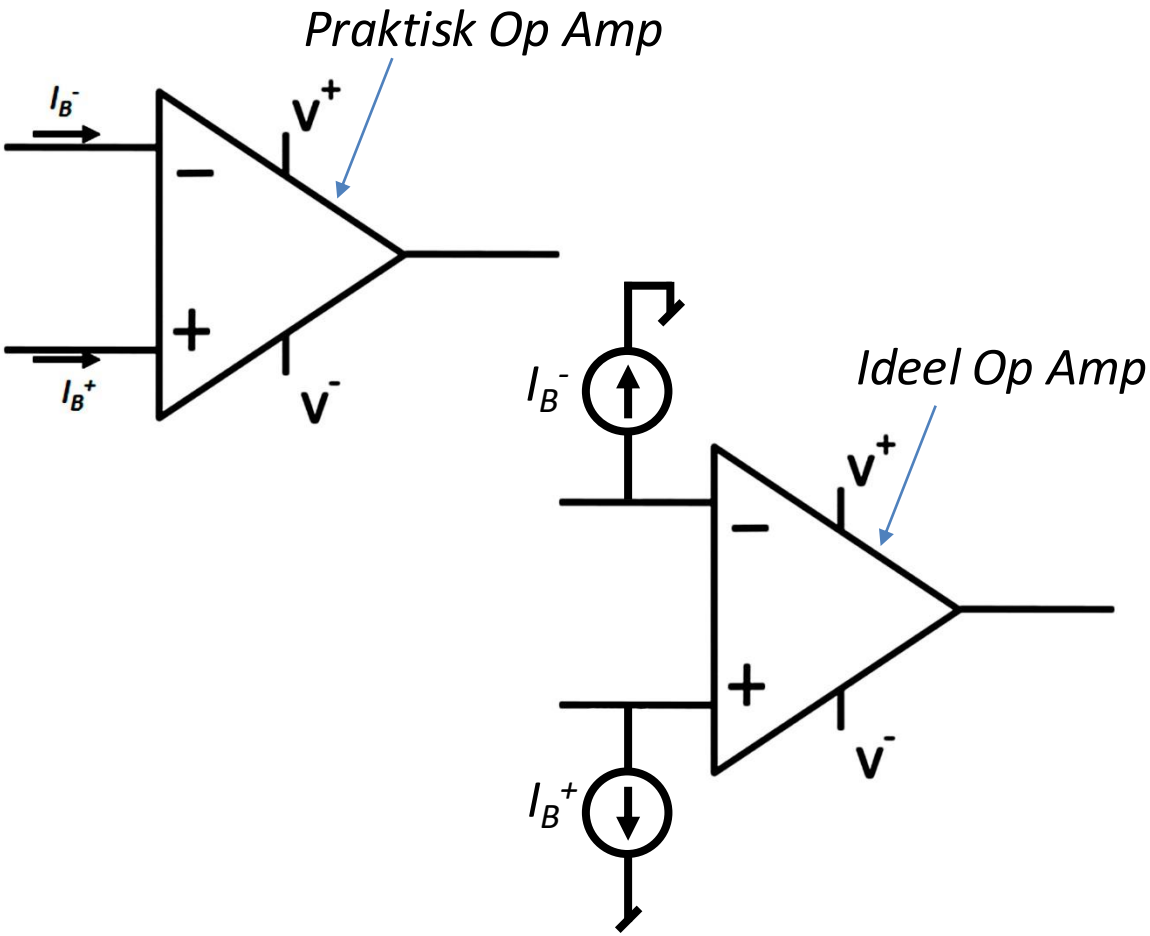
Placering af biasstrømme i fejlækvivalent



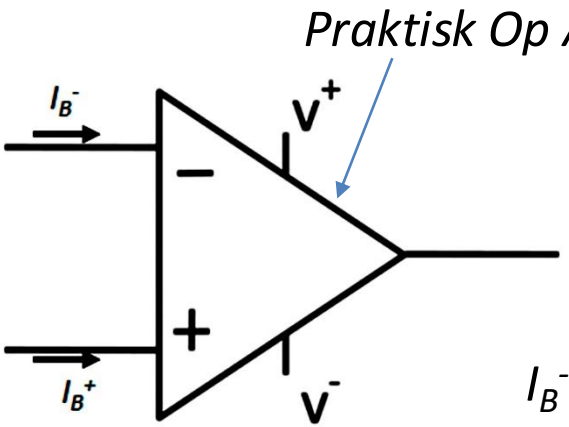
Placering af biasstrømme i fejlækvivalent

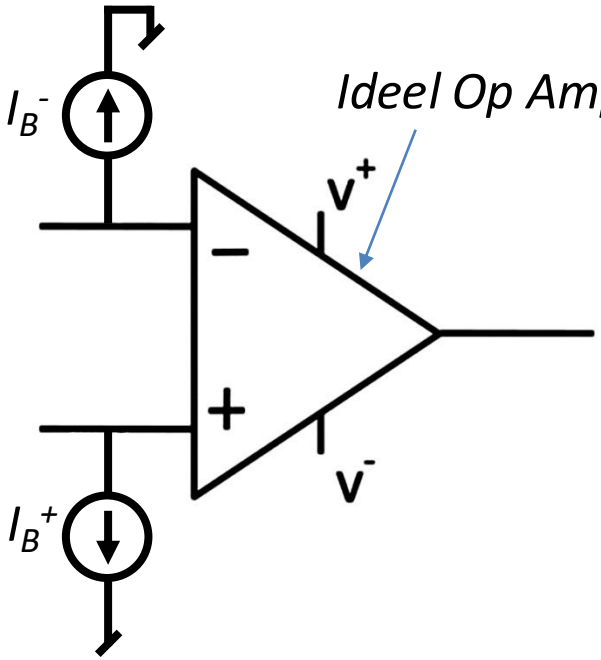


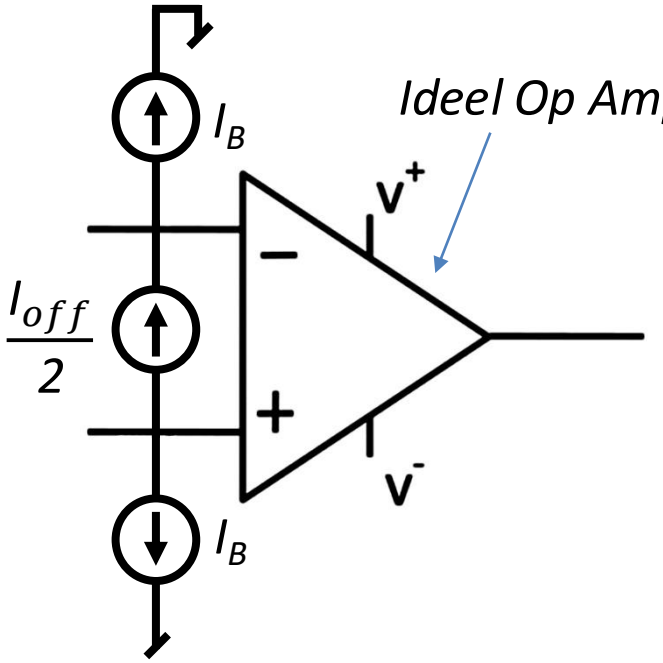
Placering af biasstrømme i fejlækvivalent



Placering af biasstrømme i fejlækvivalent







$$I_B^+ = I_B + \frac{I_{off}}{2}$$

$$I_B^- = I_B - \frac{I_{off}}{2}$$

Offsetjustering

Internt** eller **eksternt

Offsetjustering

Konstruktionsopgaven

Internt** eller **eksternt

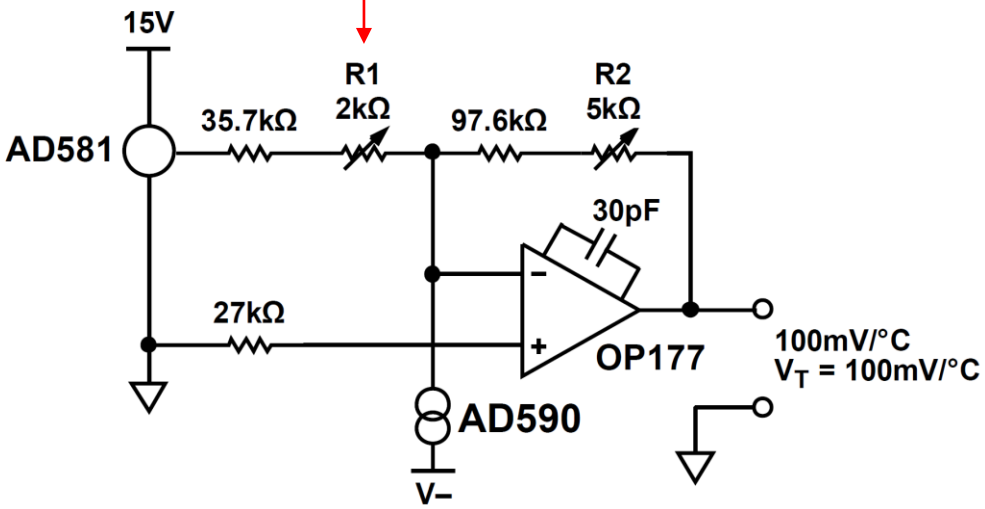


Figure 13. 2-Temperature Trim

Offsetjustering

Konstruktionsopgaven

Internt** eller **eksternt

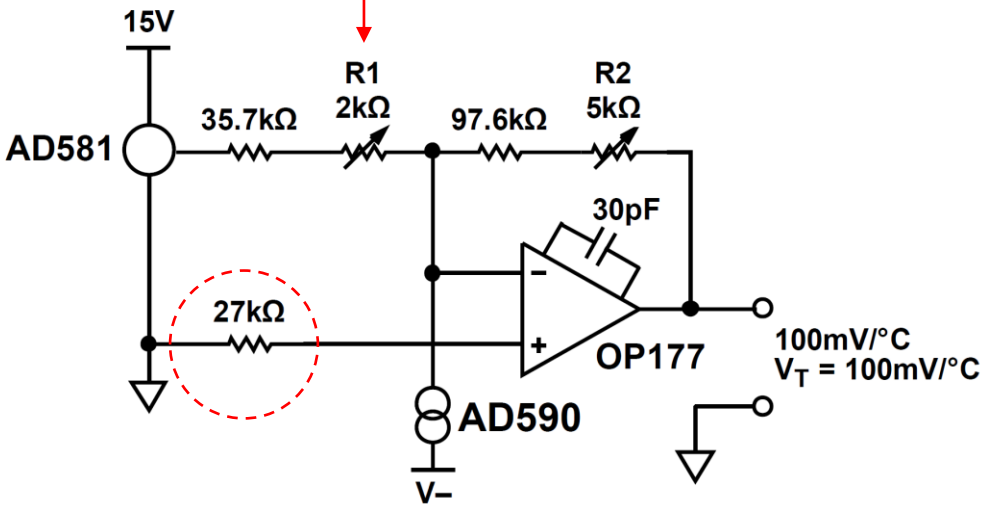


Figure 13. 2-Temperature Trim

Offsetjustering

Konstruktionsopgaven

Internt** eller **eksternt

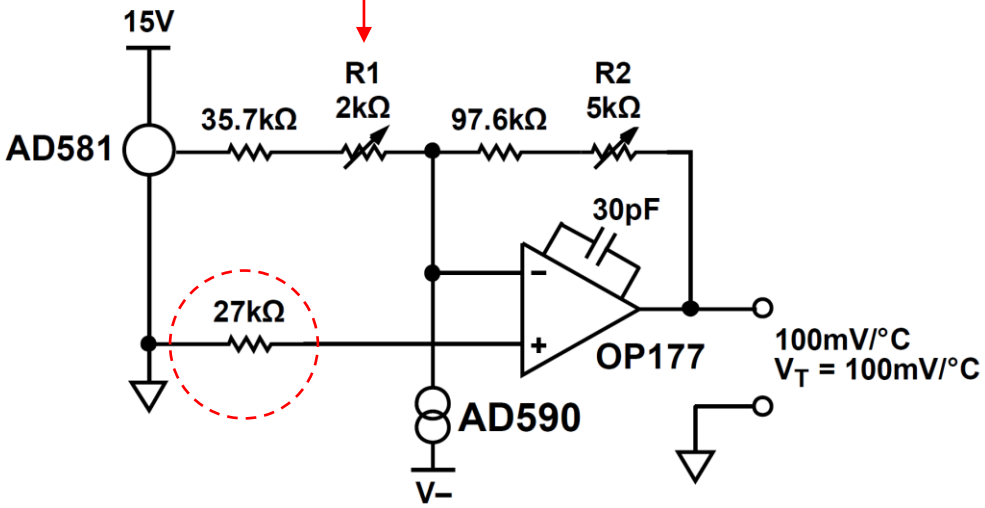


Figure 13. 2-Temperature Trim

*Man kan bortjustere **begyndelsesoffset***

Offsetjustering

Konstruktionsopgaven

Internt** eller **eksternt

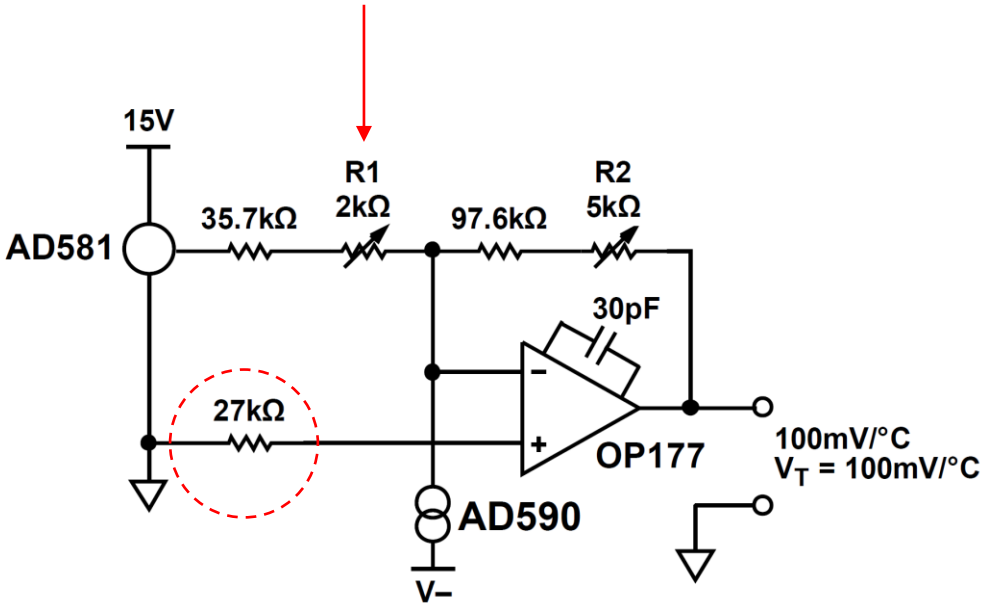


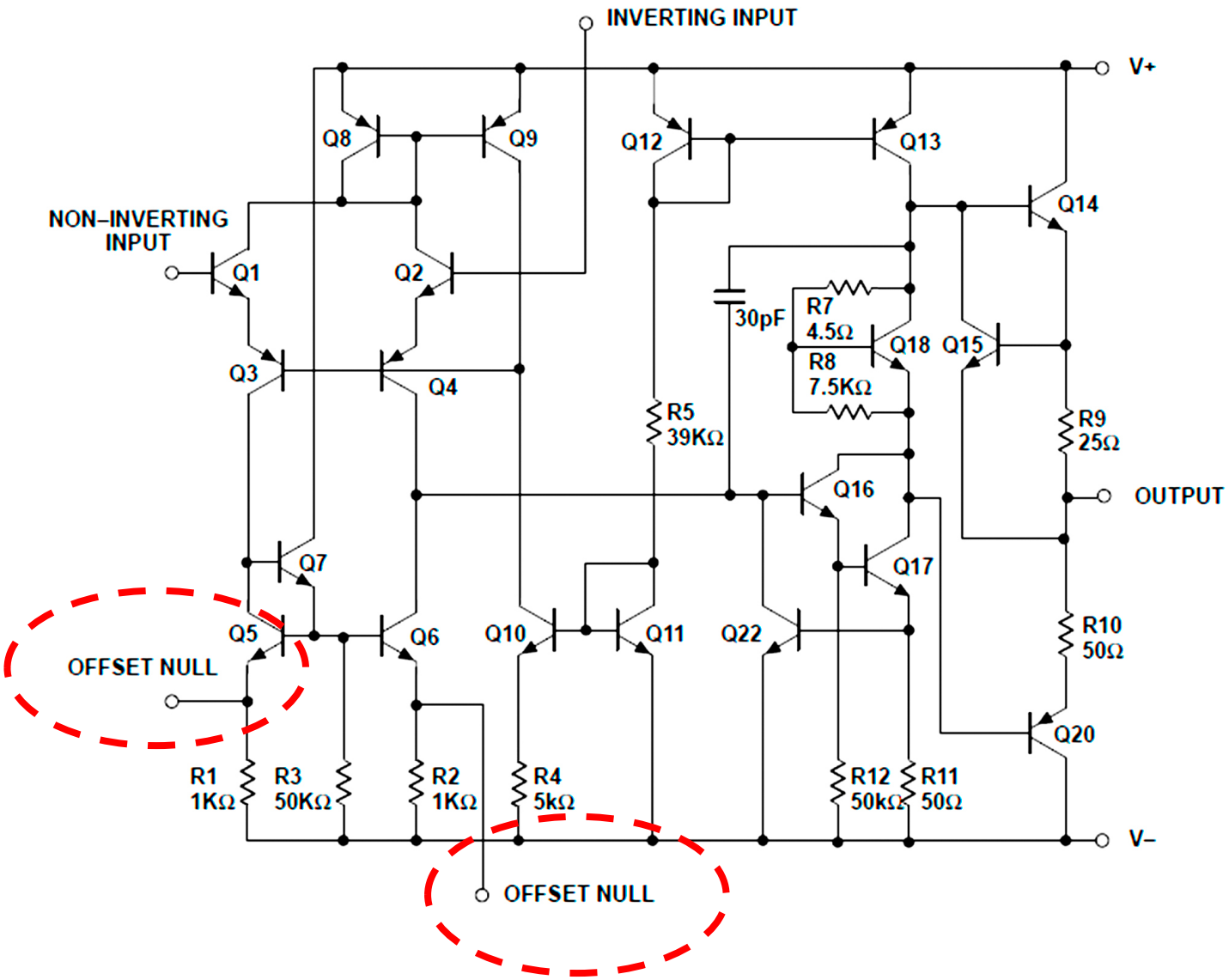
Figure 13. 2-Temperature Trim

*Man kan bortjustere **begyndelsesoffset***

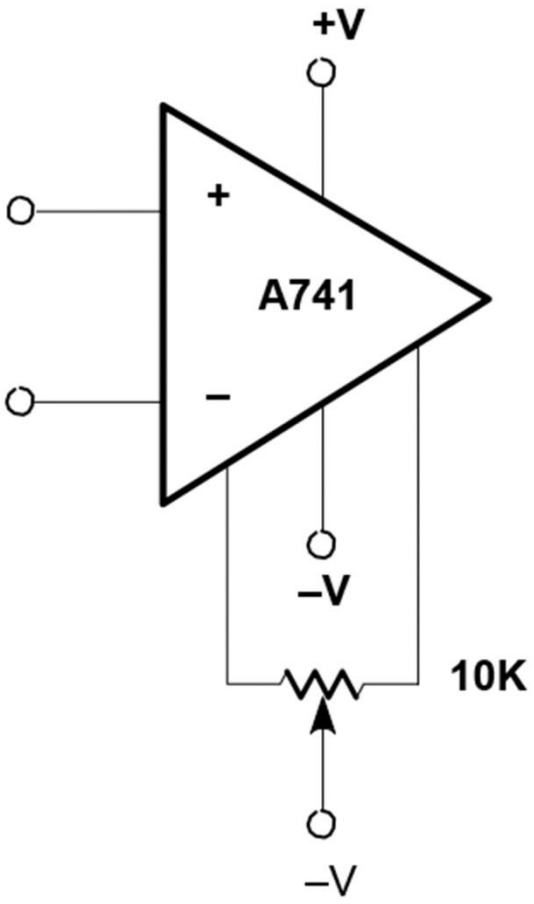
*men **ALDRIG drift** (eks. pga. temperatur)*

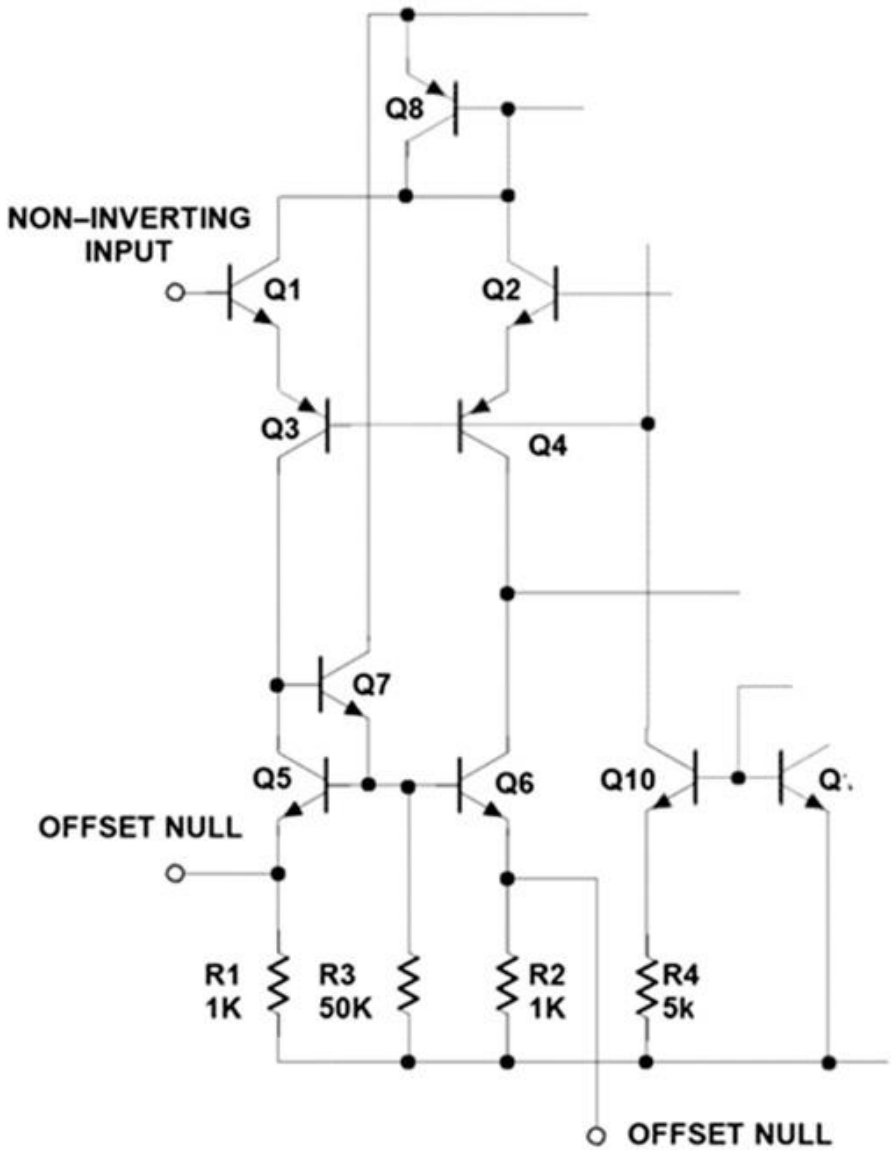


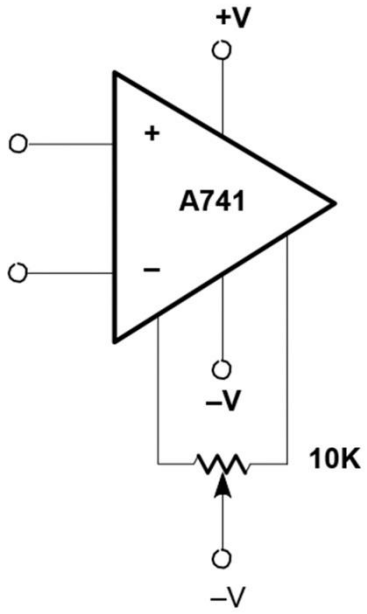
Intern offsetjustering

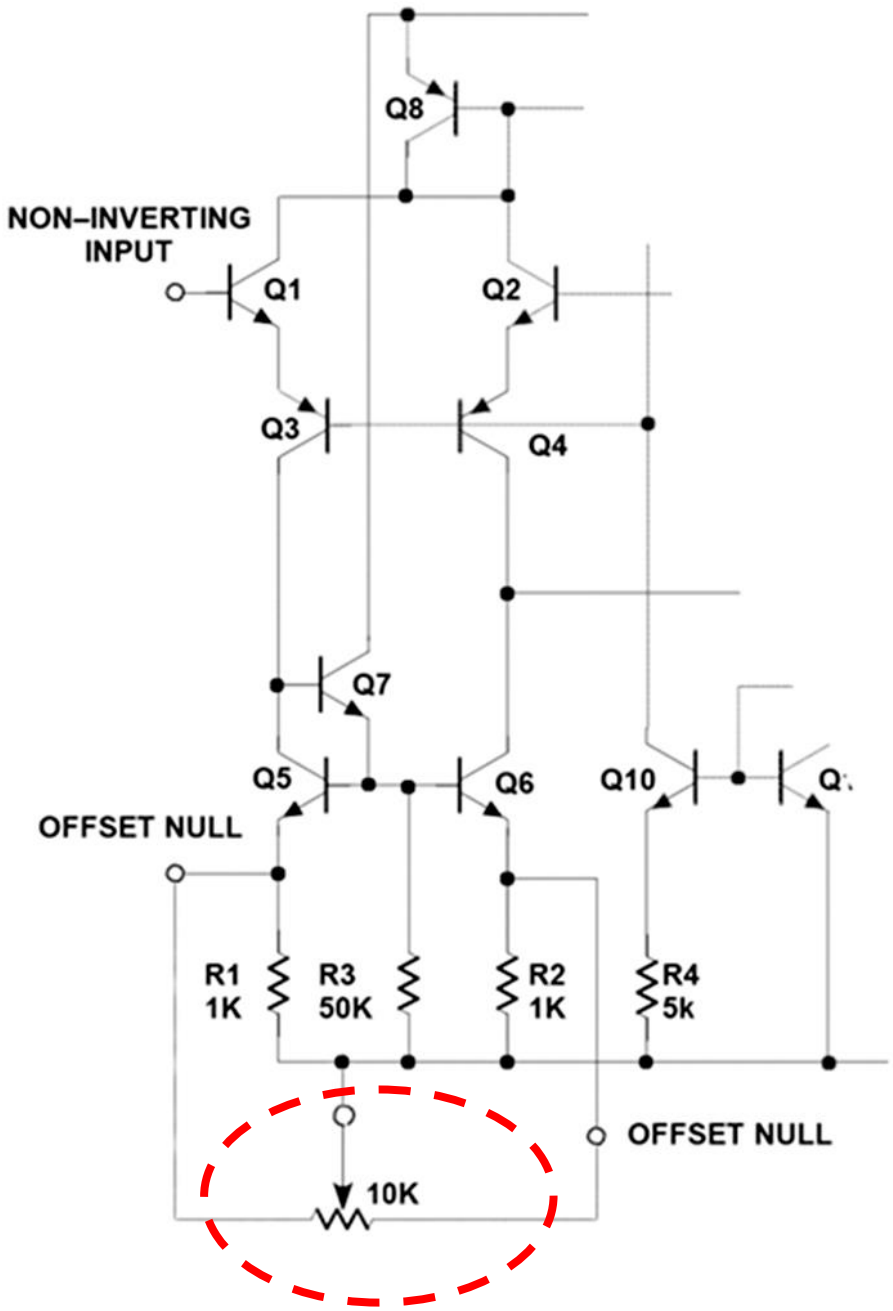


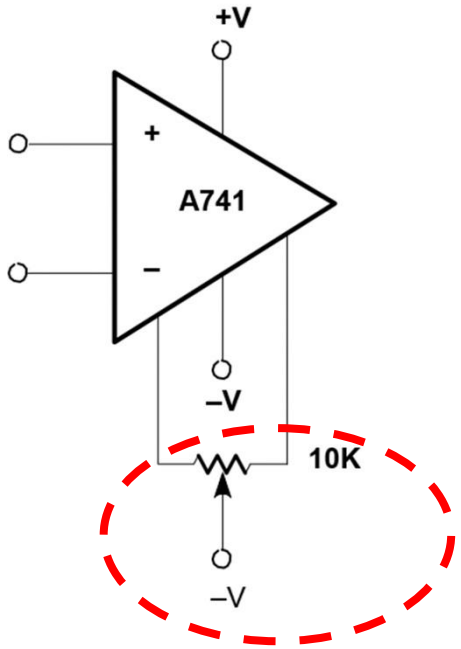
Sidste side i µA741 datablad:

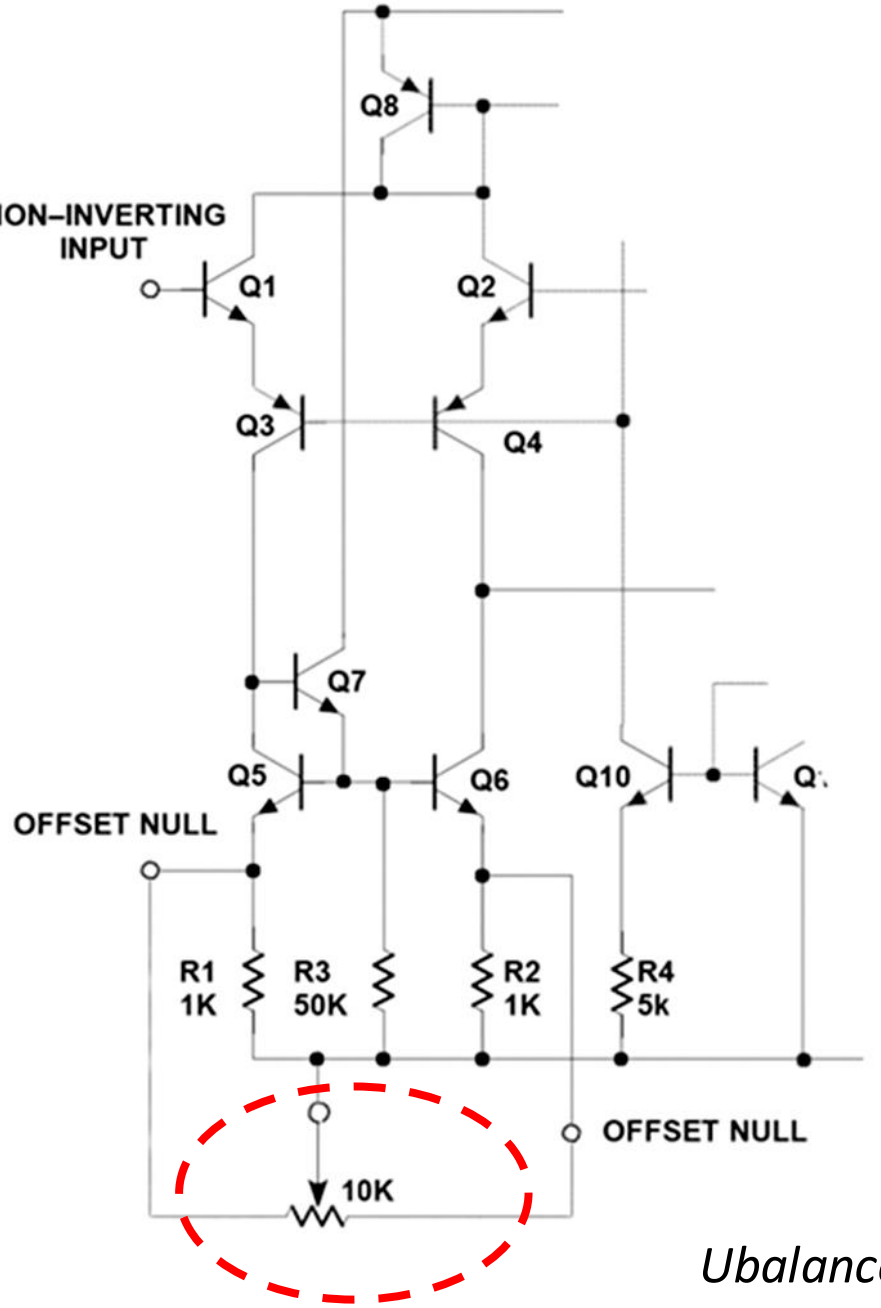


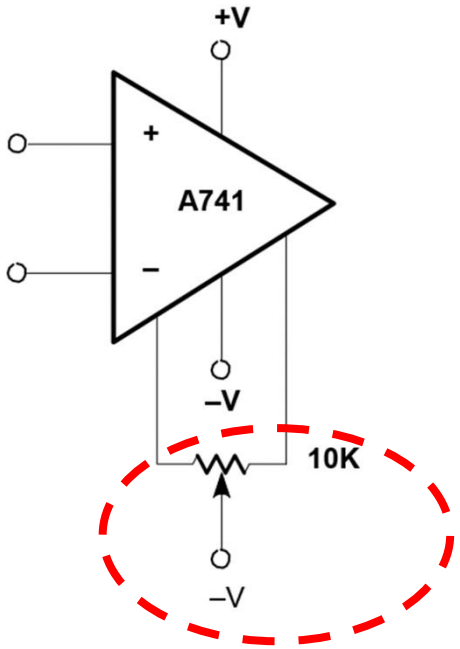

Q8
NON–INVERTING
INPUT
Q1
Q2
Q3
Q4
Q7
Q5
Q6
Q10
OFFSET NULL
R1
1K
R3
50K
R2
1K
R4
5k
OFFSET NULL


+V
+
A741
–
–V
10K
–V


Q8
NON–INVERTING
INPUT
Q1
Q2
Q3
Q4
Q7
Q5
Q6
Q10
OFFSET NULL
R1
1K
R3
50K
R2
1K
R4
5k
OFFSET NULL
10K


+V
+
A741
–
–V
10K
–V



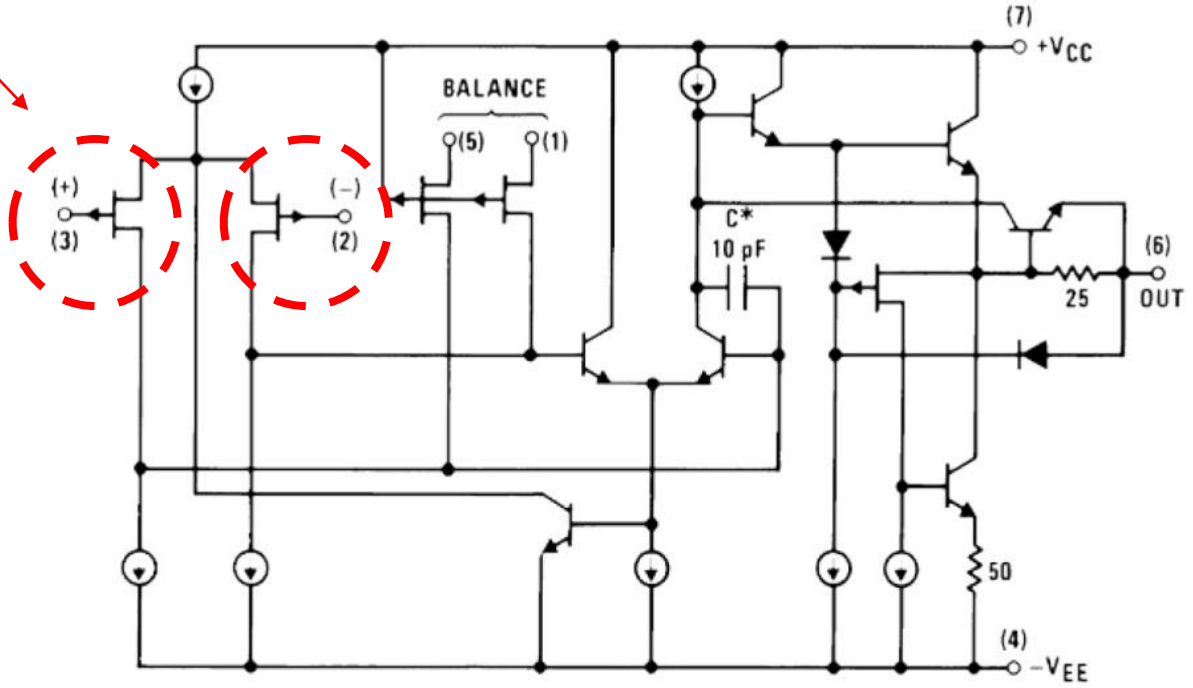


Ubalance i differenstrinnets DC-arbejdspunkter

LF355/LF356/LF357

FETs i indgangene vil sænke både I_b og I_{off}

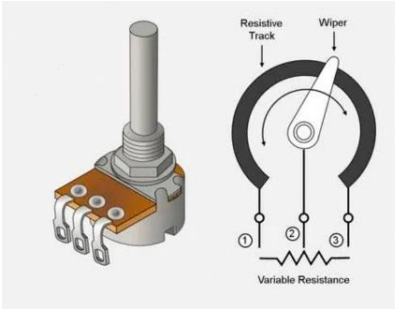
Simplified Schematic



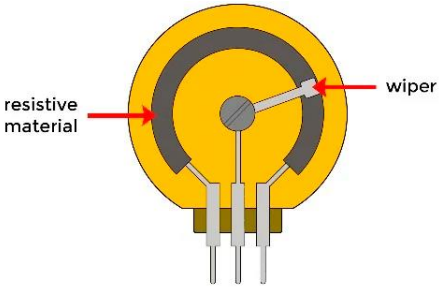
*3pF in LF357 series.

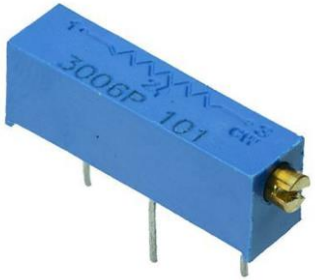
BI-FET™, BI-FET II™ are trademarks of National Semiconductor Corporation.

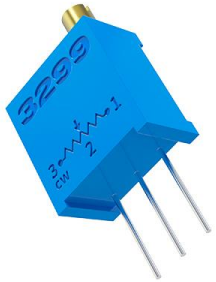
Potentiometre - og hvordan de forbindes



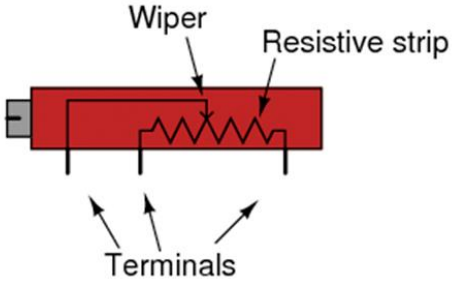








Linear potentiometer construction



Vedr. fysisk placering af AD590

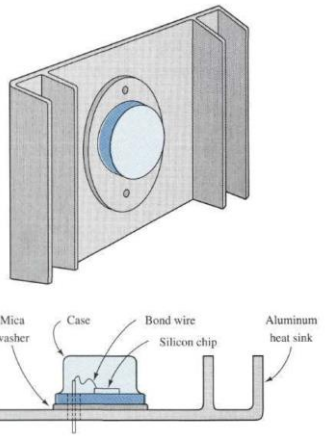


Figure 10.1
Typical heat sink for a power device.

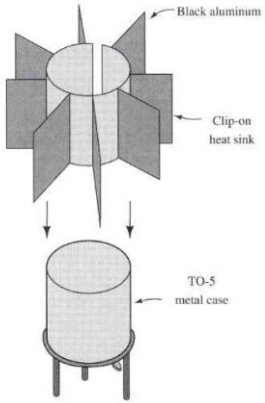


Figure 10.4
Clip-on heat sinks are suitable for power dissipations of a few watts.

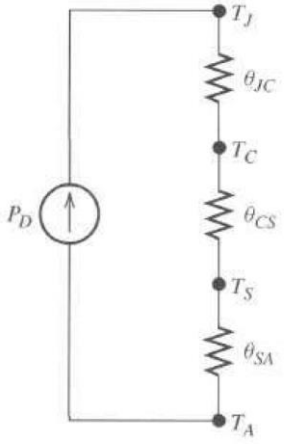


Figure 10.2
The flow of heat from junction to ambient is analogous to this series electrical circuit.

*Vurdér **hvor** I vil måle temperaturen, og om AD590 **opnår** denne temperatur.*